



Global Academy of Technology

(Affiliated to Visvesvaraya Technological University, Belagavi,
recognized by Karnataka, Approved by AICTE, New Delhi)

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Dept. of Electronics and Communication Engineering Communication Simulink Toolbox Laboratory Manual (ECE23607A)

Name: _____

USN: _____

Semester: _____ Section: _____



Global Academy of Technology

Ideal Homes Township, Rajarajeswari Nagar, Bengaluru – 560098



Certificate

This is to certify that Mr./Ms. _____
bearing USN _____ of the department of Electronics and Communication Engineering has
satisfactorily completed the course of experiments in **Communication Simulink Toolbox** Laboratory
prescribed by the Visvesvaraya Technological University, Belagavi, in the academic year 2025-26.

Marks Allotted for Practical Work			
Record	/20	CIE	/30
TOTAL		/50	

Signature of Staff

Signature of Head of the Department

Date:

Knowledge and Attitude Profile (WK)

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behaviour and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Outcomes (POs)

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyse complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Educational Objectives (PEOs)

Electronics and Communication Engineering graduates will be able to:

PEO1: Acquire technical knowledge of fundamentals of basic sciences, Electronics and Communication Engineering and applied mathematics for the rapid changes in technology.

PEO2: Analyze, design, apply, evaluate and solve complex problems in technology and management through engineering skills.

PEO3: Demonstrate team spirit by applying ethics and professional behaviour.

PEO4: Involve in research and development in the field of electronics and communication engineering and multi-disciplinary areas for lifelong learning.

Program Specific Outcomes (PSOs)

After successful completion of Electronics and Communication Engineering Program, the graduates will be able to:

PSO1: Analyze, design and implement emerging electronics and communication systems using devices, subsystems and networking.

PSO2: Exhibit technical skills necessary to choose careers in the design, implement and testing, management and operation of Electronics and Communication circuits and systems.

SYLLABUS

Communication Simulink Toolbox Laboratory (ECE23607A)

Subject Code	ECE23607A	CIE Marks	50
Hours/Week (L: T: P)	0:0:2	SEE Marks	50
Credits	1	Examination Hours	03

Course Learning Objectives: Student will be taught:

CLO1	Simulate discrete time signals.
CLO2	DPSK and QPSK Transmitter and Receiver.
CLO3	Random noise generation histogram plotting.
CLO4	OFDM modulation and demodulation.
CLO5	Filtering operations.

Course Outcome	Description	Mapped POs/PSOs	Cognitive Level	WK	Class Hours
CO1	Generate Sine, Square, Triangular and Ramp signals and display waveforms.	PO1, PO2, PO7, PO8, PO11, PSO2	L3	WK4, WK6, WK9	6
CO2	Perform DPSK and QPSK modulation and demodulation in presence of noise.	PO1, PO2, PO7, PO8, PO11, PSO2	L3	WK4, WK6, WK9	6
CO3	Generate random noise and plot histogram for various types of noise.	PO1, PO2, PO7, PO8, PO11, PSO2	L3	WK4, WK6, WK9	6
CO4	Perform various operations on OFDM.	PO1, PO2, PO7, PO8, PO11, PSO2	L3	WK4, WK6, WK9	5
CO5	Demonstrate different filtering operations on signals.	PO1, PO2, PO7, PO8, PO11, PSO2	L3	WK4, WK6, WK9	5

List of experiments to be conducted using simulation Tool

Sl. No.	Experiments	RBT Level
1	Generation of signals such as Sine, Square, Triangular and Ramp.	L4
2	DPSK Transmitter and Receiver in Simulink.	L4
3	Generate a multiband signal using the Communications Toolbox.	L4
4	Random noise generation using Simulink & display histogram plots of Gaussian and Uniform noise.	L4
5	Perform Equalization, Convolution, and Cyclic Prefix Addition on basic OFDM.	L4
6	QPSK Transmitter and Receiver in Simulink	L4
7	Design and implementation of Low pass and High pass filter.	L4
8	Design and implementation of Band pass and Band reject filter.	L4
9	Obtain the scatter plots & eye diagrams of a QPSK signal to visualize the signal behaviour in presence of AWGN.	L4
10	Perform OFDM modulation and demodulation.	L4

Textbooks:

1. W. Y. Yang, Y. S. Cho, J. W. Lee, W. G. Jeon, J. H. Paik, J. K. Kim, M. H. Lee, K. I. Lee, K. W. Park, and K. S. Woo, MATLAB/Simulink for Digital Communication, A-Jin Publishing Co., Ltd., 2009.
2. K. C. Raveendranathan, Communication Systems Modelling and Simulation using MATLAB and Simulink, 1st ed., Routledge, 2011.

Scheme of Examination

Continuous Internal Evaluation (CIE):

As part of CIE process, progressive continuous evaluation is done for laboratory work on weekly basis of conduct of experiment by student either individually or in group based on the laboratory. The breakup of the marks allocated is given in the TABLE-1.

TABLE-1 WEEKLY EVALUATION OF CONDUCTION OF EXPERIMENT		
SL. NO.	ACTIVITY	MAX MARKS
1	Conduction of experiment and documentation	10
2	Analysis & interpretation of results	5
3	Viva voce	5
TOTAL		20

Internal exam conducted at the end of the semester or on completion of a predefined set of experiments based on the laboratory. The evaluation detail of laboratory internal exam is given in TABLE-2.

TABLE-2 LAB INTERNAL EXAM		
SL. NO.	ACTIVITY	MAX MARKS
1	Detailed write-up about the experiment with relevant procedure and calculation	5
2	Conduction of experiment	20
3	Viva voce	5
TOTAL		30

TABLE-3 FINAL CIE CALCULATION		
SL. NO.	METRICS USED	MAX MARKS
1	Average of all weekly evaluation of conduct of experiment	20
2	LAB Internal Exam	30
TOTAL		50

Semester End Examination (SEE):

- All laboratory experiments are to be included for practical examination.
- Students can pick one experiment from the questions lot prepared by the examiners.
- Change of experiment is allowed only once and 20% Marks allotted to the conduction part to be made zero.

Semester End Examination Evaluation		
SL. NO.	ACTIVITY	MARKS
1	Write-up	15
2	Conduction	70
3	Viva Voce	15
TOTAL		100

Note: The marks scored will be proportionately reduced to 50

CO-PO and PSO Mapping													
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	-	-	-	1	1	-	-	1	-	-	1
CO2	3	2	-	-	-	1	1	-	-	1	-	-	1
CO3	3	2	-	-	-	1	1	-	-	1	-	-	1
CO4	3	2	-	-	-	1	1	-	-	1	-	-	1
CO5	3	2	-	-	-	1	1	-	-	1	-	-	1
Average	3	2	-	-	-	1	1	-	-	1	-	-	1

Low-1: Medium-2: High-3

Evaluation of Lab Marks-RUBRICS

Sl. No.	Experiment	Engineering knowledge on Communication	Conduction of experiment	Apply the knowledge of Engineering to Communication	Calculation and Results	The student correctly written and answers for viva questions	Total	Signature of the faulty with date
	Weightage	Weightage (1)	Weightage (2)	Weightage (1)	Weightage (1)	Weightage (1)	Marks	
Max Marks		(Score x Wtg) Max score = 1	Max score = 3	Max score = 3	Max score = 5	Max score = 5	20	
1	Generation of signals such as Sine, Square, Triangular and Ramp.							
2	DPSK Transmitter and Receiver in Simulink.							
3	Generate a multiband signal using the Communications Toolbox.							
4	Random noise generation using Simulink & display histogram plots of Gaussian and Uniform noise.							
5	Perform Equalization, Convolution, and Cyclic Prefix Addition on basic OFDM.							
6	QPSK Transmitter and Receiver in Simulink.							
7	Design and implementation of Low pass and High pass filter.							
8	Design and implementation of Band pass and Band reject filter.							
9	Obtain the scatter plots & eye diagrams of a QPSK signal to visualize the signal behavior in presence of AWGN.							
10	Perform OFDM modulation and demodulation.							
Average (A) = 20								
Internal Assessment Marks (CBCS)) (B) = 30								
Total Marks (A + B) = 50								

Lab manuals and observation assessment on weekly basis (Max Marks 20)

Performance	PO/PSO mapped	Exceptional (score3)	Satisfactory (score2)	Marginal (score1)	Marks (Score*Wtg)
Engineering knowledge on Communication Weightage (1)	PO1, PO2, PO7, PO8, PO11, PSO2	Effectively applies principles of fundamental engineering to design Simulink model	Effectively applies principles of fundamental engineering to design Simulink model with minor errors	Does not know how to apply principles of fundamental engineering knowledge to design Simulink model	1
Conduction of experiment Weightage (2)	PO1, PO2, PO7, PO8, PO11, PSO2	Ability to demonstrate, conduction of experiment by applying engineering knowledge	Able to demonstrate conduction of experiment by applying engineering concepts to design Simulink model with minor errors	The student does not exhibit ability to conduct experiment by invoking the tool applying engineering skills to design Simulink model	6
Apply the knowledge of engineering to Communication Weightage (1)	PO1, PO2, PO7, PO8, PO11, PSO2	Effectively apply the engineering knowledge to design Simulink model	Able to apply the engineering knowledge to design Simulink model with minor errors	Does not know how to apply the engineering knowledge to design Simulink model	3
Calculation and Results Weightage (1)	PO1, PO2, PO7, PO8, PO11, PSO2	Effectively evaluate the model and obtain simulated output waveforms	Provide inputs to given model to obtain simulated output waveforms	Does not know how to apply inputs and how to start with the tool for implementation of a Simulink model	5
The student correctly written and answers for viva questions Weightage (1)	PO1, PO2, PO7, PO8, PO11, PSO2	Answers the viva questions By using approach of concept	The student answered only some of the viva questions	The student has not written and answered for the viva questions	5

Guidelines

- Student assessed on his/her participation and performance in conduction of lab, inweekly basis.
- The average value considered as final marks for the record/manuals.
- Late comers (after10 mins) are not allowed to the laboratory.
- Arrive prepared with experiment understanding and get instructor approval before powering circuits.
- Tidy workspace, return equipment/components, and unplug devices after use.
- Avoid mobile phones, eating/drinking, or unauthorized experiments.
- Wear closed-toe shoes to prevents foot injuries, no sandals or open footwear.
- Refrain from handling equipment without permission or leaving powered setups unattended.
- The student required to wear decent dress code.

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INTRODUCTION TO MATLAB SIMULINK

Simulink is a software for modelling, simulating, and analysing dynamical systems. It supports linear and nonlinear systems, modelled in continuous time, sampled time or hybrid of two. Systems can also be multirate, i.e., have different parts that are sampled or updated at different rates.

For modelling, Simulink provides a graphical user interface (GUI) for building models as block diagrams, using click-and-drag mouse operations. With this interface, you can draw the models just as you would with pencil and paper (or as most textbooks depict them). Simulink includes a comprehensive block library of sinks, sources, linear and nonlinear components, and connectors. You can also customize and create your own blocks.

Models are hierarchical. This approach provides insight into how a model is organized and how its parts interact. After you define a model, you can simulate it, using a choice of integration methods, either from the Simulink menus or by entering commands in MATLAB's command window. The menus are particularly convenient for interactive work, while the command-line approach is very useful for running a batch of simulations.

Using scopes and other display blocks, you can see the simulation results while the simulation is running. In addition, you can change parameters and immediately see what happens, for "what if" exploration. The simulation results can be put in the MATLAB workspace for post processing and visualization. And because MATLAB and Simulink are integrated, you can simulate, analyze, and revise your models in either environment at any point.

Simulink is a block diagram environment for multi-domain simulation and Model-Based Design. It supports simulation, automatic code generation, and continuous test and verification of embedded systems. Simulate dynamic systems leveraging graphical editors, customizable block libraries, and solvers for modeling.

You can use Simulink to model a system and then simulate the dynamic behavior of that system. The basic techniques you use to create the simple model in this tutorial are the same techniques that you use for more complex models.

To create this simple model, you need four Simulink blocks. Blocks are the model elements that define the mathematics of a system and provide input signals:

How to use Simulink

Step 1: Click "Start" button of MATLAB. Search for the "Simulink" option and click "Library Browser" option.

1.1 Open Simulink

1. First open MATLAB from the install directory or desktop shortcut.
2. Once open locate the SIMULINK icon as show in Figure 1 or simply type "simulink" in the command window.

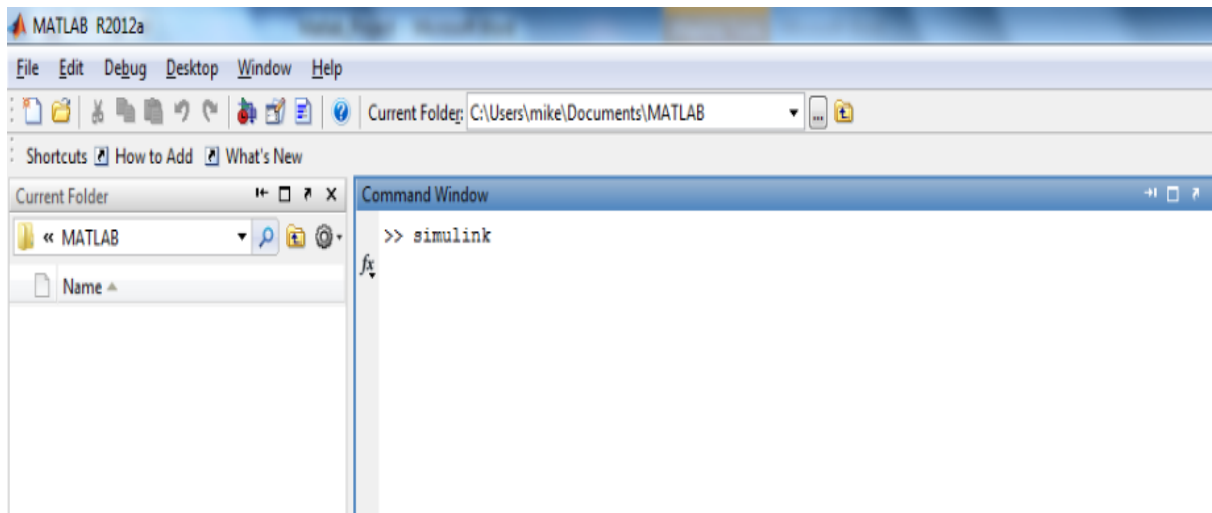


Figure 1: MATLAB

Step 2: Select "File" button ! "New" ! Model

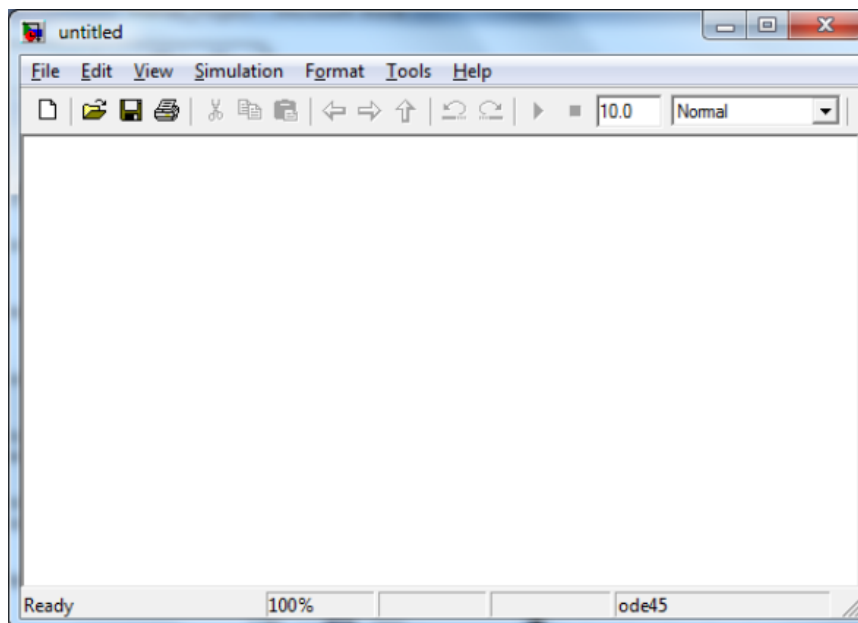


Figure 2: New Model

3. Once you open SIMULINK you will see the “Library Browser” as in Figure 3.

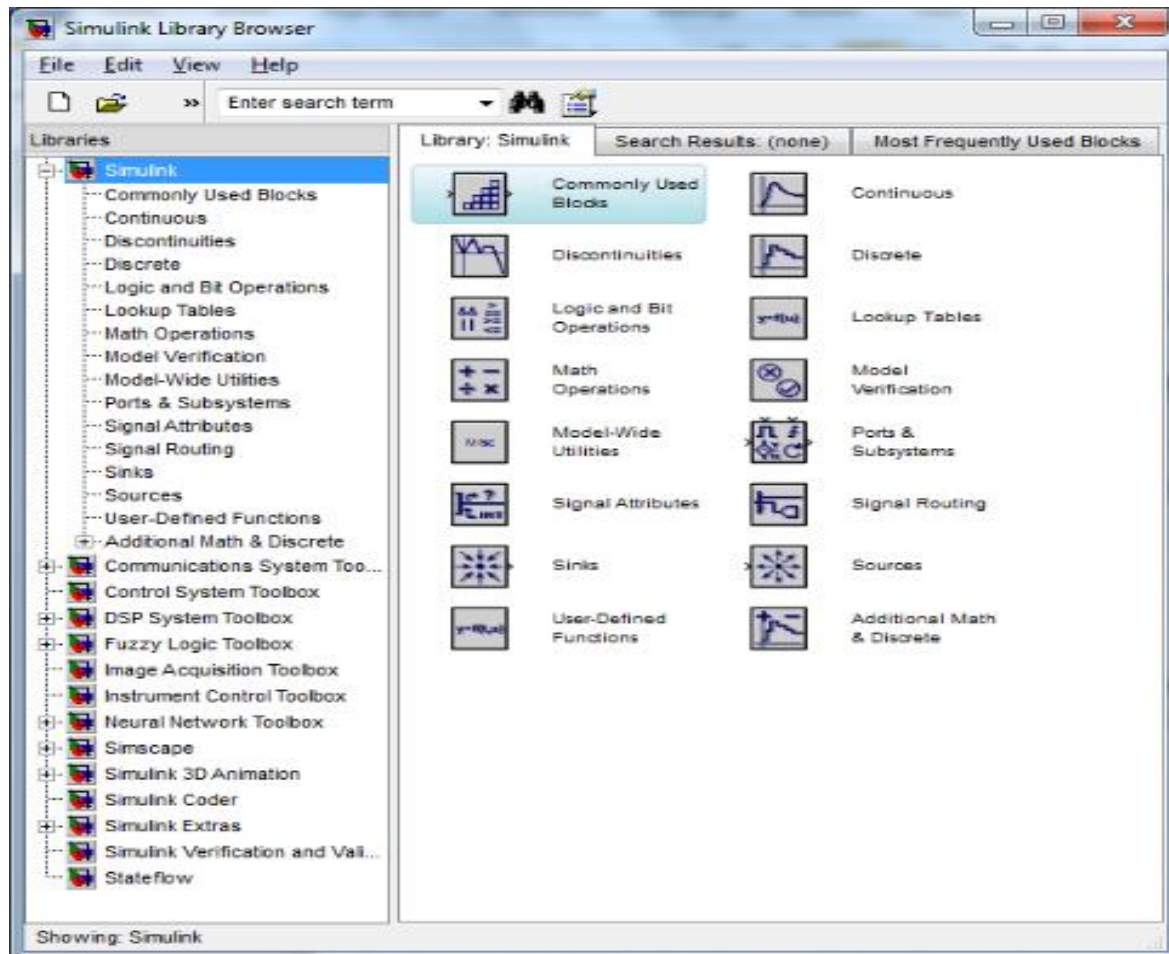


Figure 3: Library Browser

- Highlighted in Figure 2, locate and select the new model icon and a new model shown in Figure 3 will open. Once opened **SAVE** your model.

1.2 Summary

In this section we launched SIMULINK and opened a new model. We are now ready to build simulations.

2 Adding Blocks to a New Model

In the previous section we opened a new model from the “Library Browser” we will now add some blocks and run a simulation.

2.1 Adding a Sine Wave Generator

- In Figure 4 locate the “Sine Wave” block under Simulink->Sources and drag it into your new model window as seen in Figure 5.
- Under Simulink->Sinks locate the “Scope” Block and drag that to the new model window as well.
- By placing the mouse over the output of the “Sine Wave” the mouse cursor will change from an arrow to crosshairs. Left click and drag the wire from the “Sine Wave” block to the input of the “Scope”.

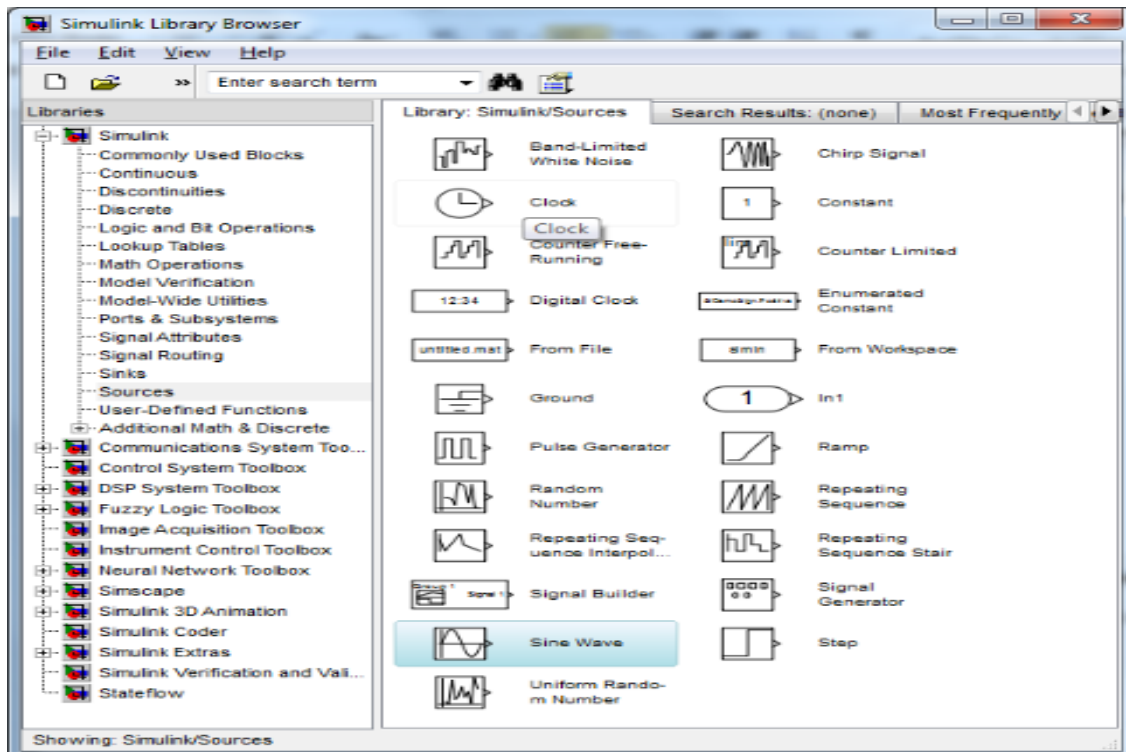


Figure 4: Sin Wave Block Located in Library Browser

Step 3: Type "sine wave" in the search blank ! drag the "sine wave" block to the model window.

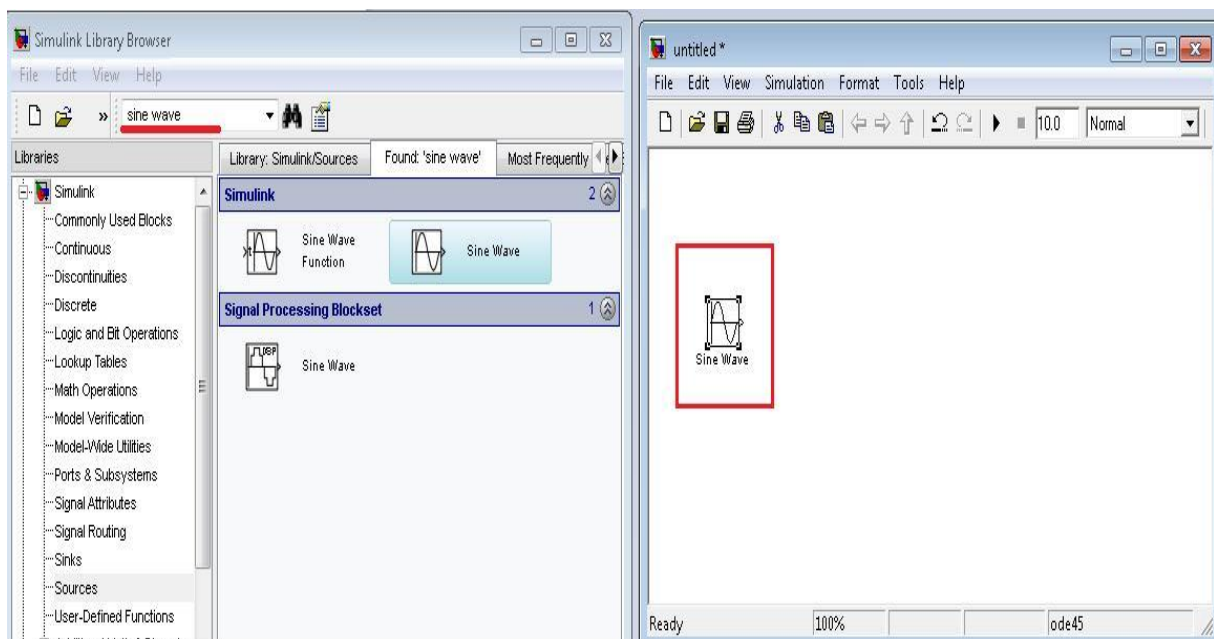


Figure 5: GUI interface

Step 4: Double click "sine wave" block ! modify the Frequency as " $500 \times 2 \times \pi$ " and Phase " $\pi/2$ ". (i.e., a sine wave with $\pi/2$ phase shift and frequency 500Hz) as shown in Figure 6.

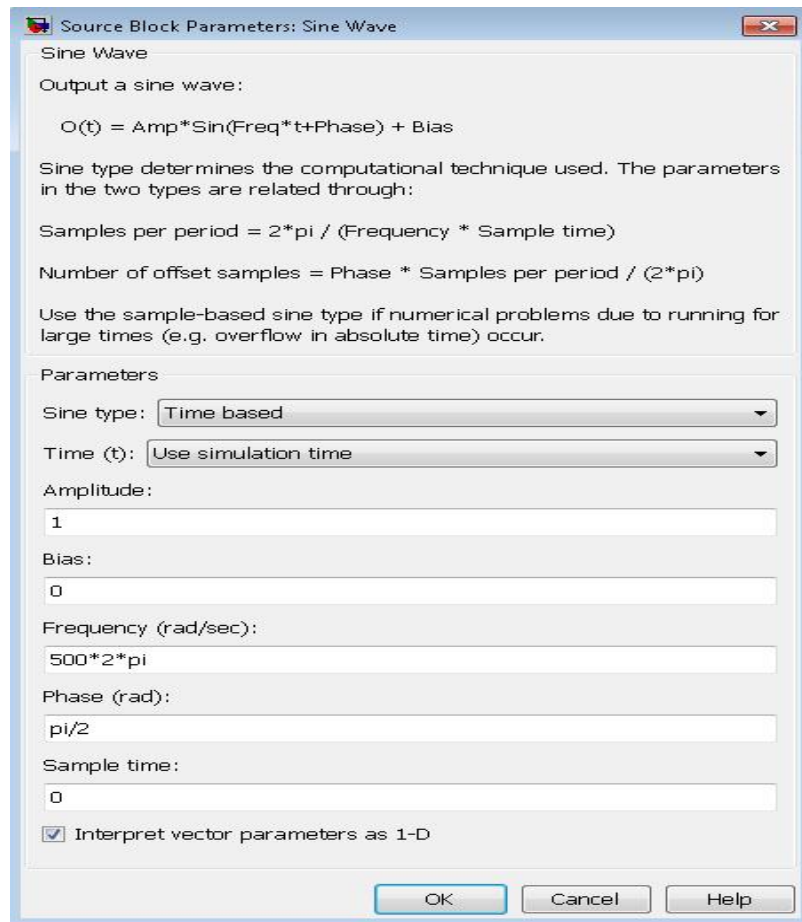


Figure 6: GUI interface - Sine Wave Setup

Step 5: Type "scope" in the search blank ! drag the "scope" block to the model window ! save model as "Model" as shown in Figure 7.

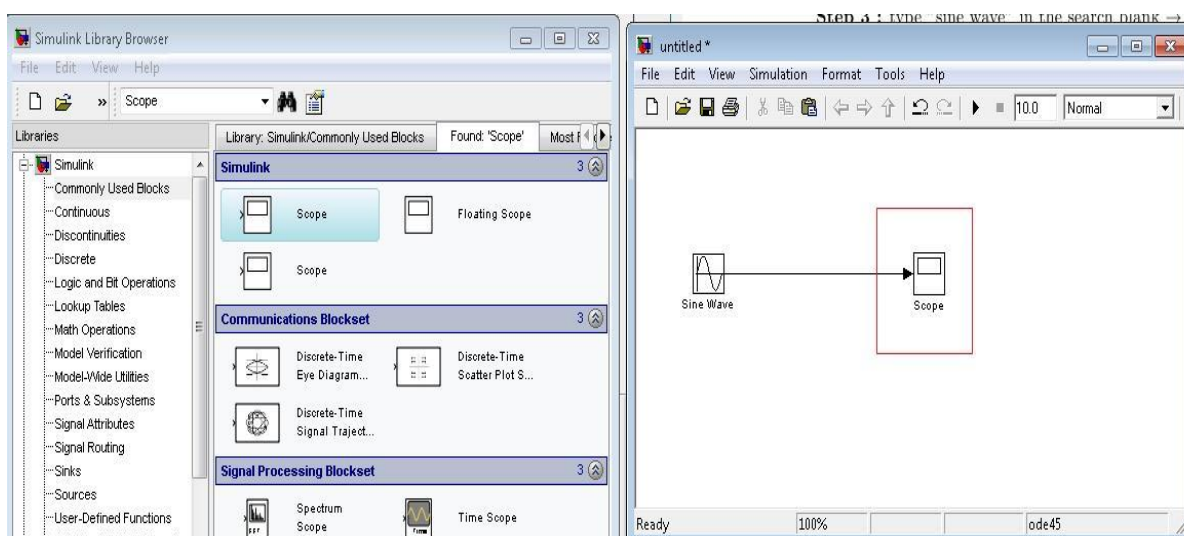


Figure 7: "Sine Wave" and "Scope" Block Added to Model

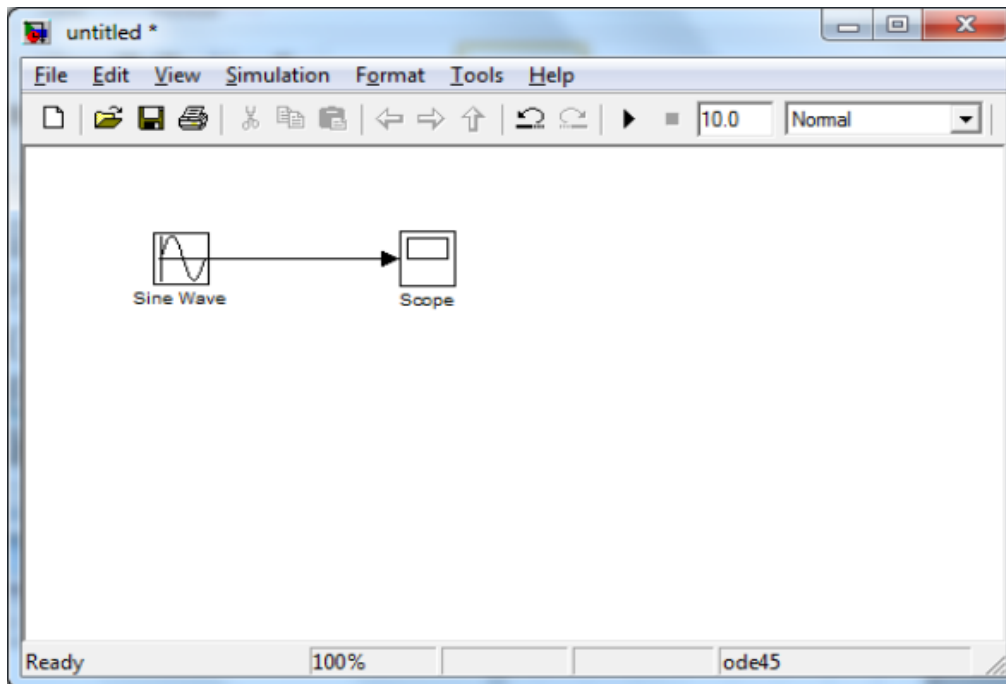


Figure 8: “Sine Wave” Connected to Scope

- Once the two blocks have been wired as shown in Figure 8, double click on the “Sine Wave” Block and make the changes as seen in Figure 9.

Note: The sample time is set to $1/10000$ or $100\mu\text{s}$ which means that the highest frequency sine wave we can create is 5000Hz and still obey the Nyquist Criteria. Play with this. Change the frequency and observe the results. You will notice that the closer you get to 5000Hz the choppier the waveform will get. Try changing the sampling rate and observe the results.

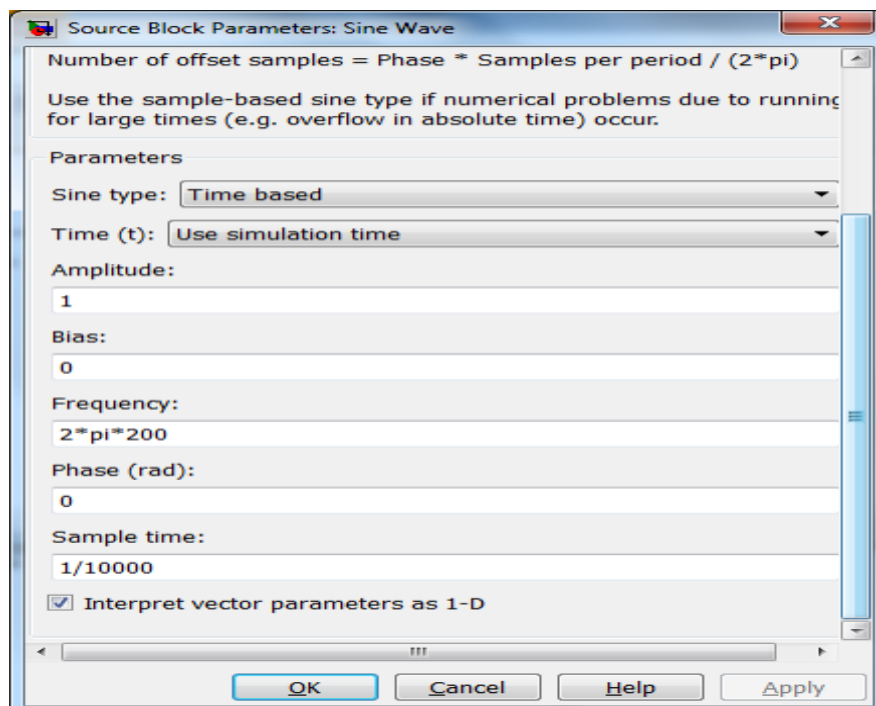


Figure 9: Change Frequency to 200Hz and the Sample Time

5. Once we have finished click “Ok” change the simulation time to 0.01secs and run the model by clicking run as seen in Figure 10.

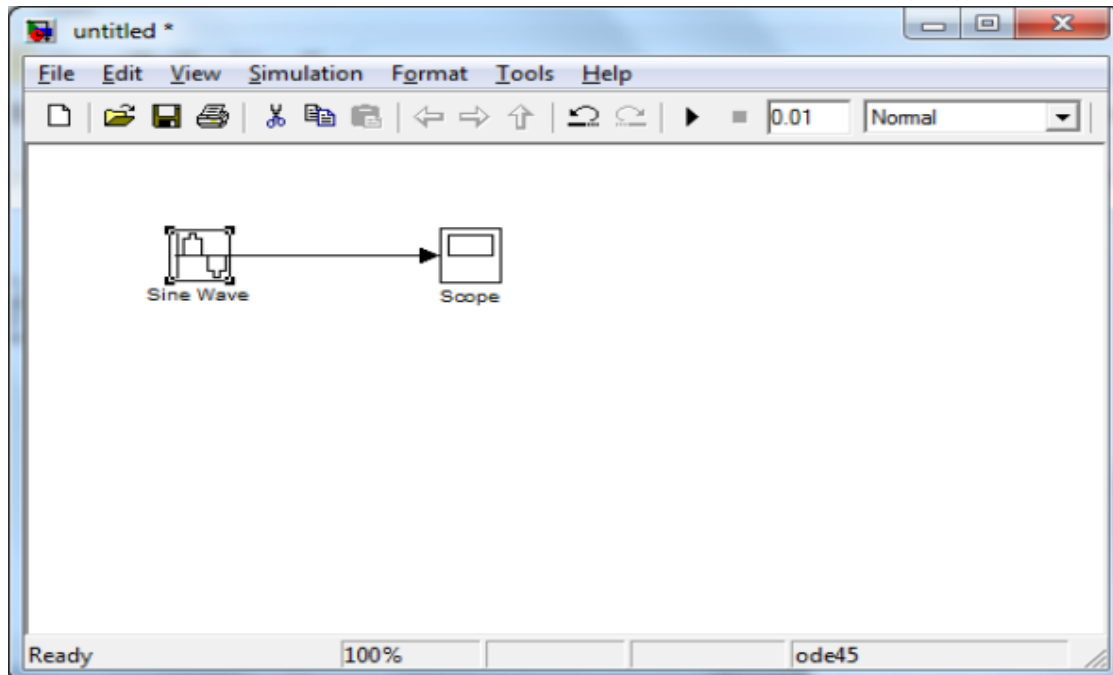


Figure 10: Click “Run”

6. Once the model has finished running double click the scope and zoom in on the plot as seen in Figure 11.

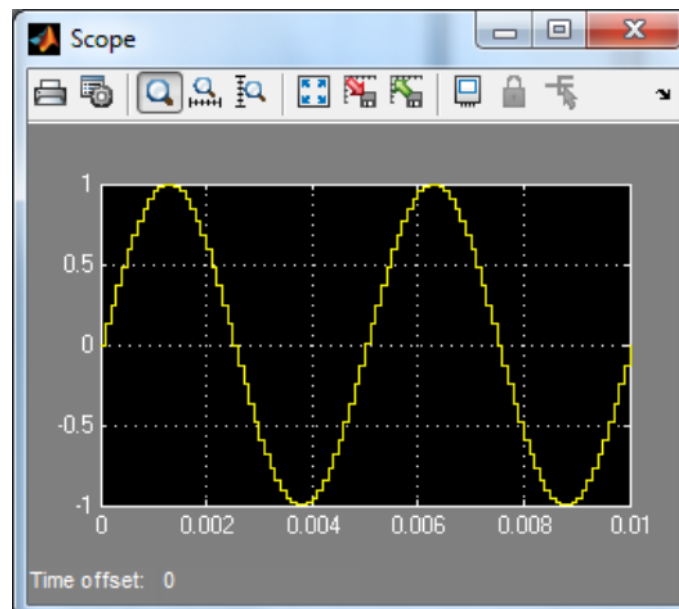


Figure 11: “Scope Plot”

2.2 Add another “Sine Wave” block at 600MHz.

Click on settings as seen in Figure 12 for the scope plot and add another axis changing the “1” to a “2”. Click “Ok”. Another input will appear on the scope. Wire the “Sine Wave” block you just added to the scope and run the simulation again. You will get two plots both containing a sine wave.

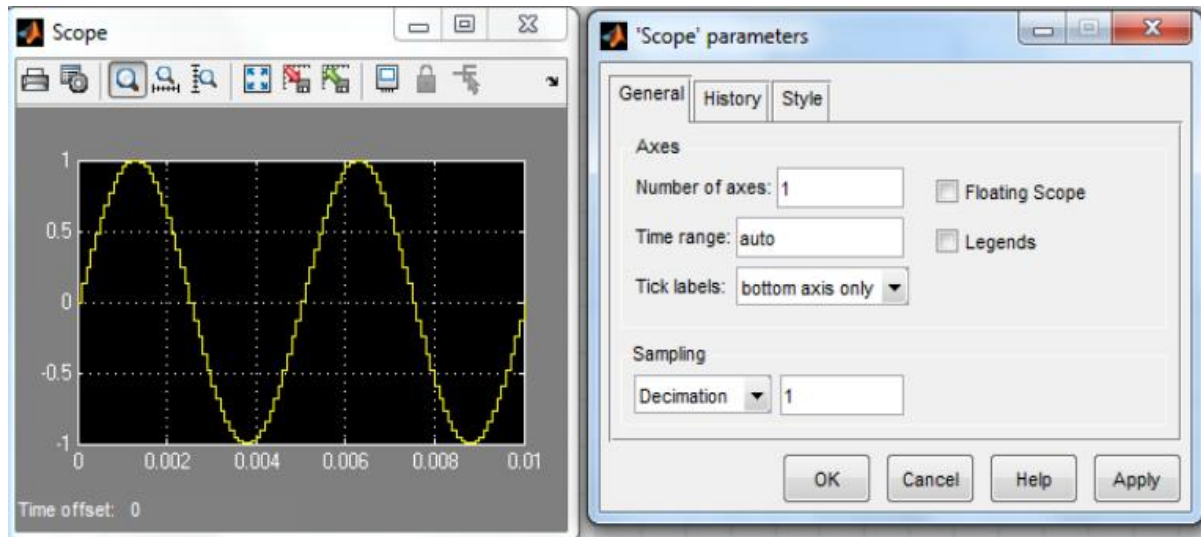


Figure 12: Adding Another Axis to the Scope Plot

2.3 Addition of sine waves

In this exercise sum two sine waves together and observe the output. Your model should look like that in Figure 13. Sum block can be found in Simulink->Commonly Used Blocks or Simulink->Math Operators. You need to add one more axis to the scope plot. Also, you need to tap off existing wires to create new connections. To do this, place the mouse over the wire and **right-click**. While right-clicking drag the wire to the new scope inputs you have created. Now try the same thing again, but this time use the “product” (multiplication) block.

Note: “Sum” and “Add” have the same overall function and the only difference is the block image that will appear when you add the block to your model.

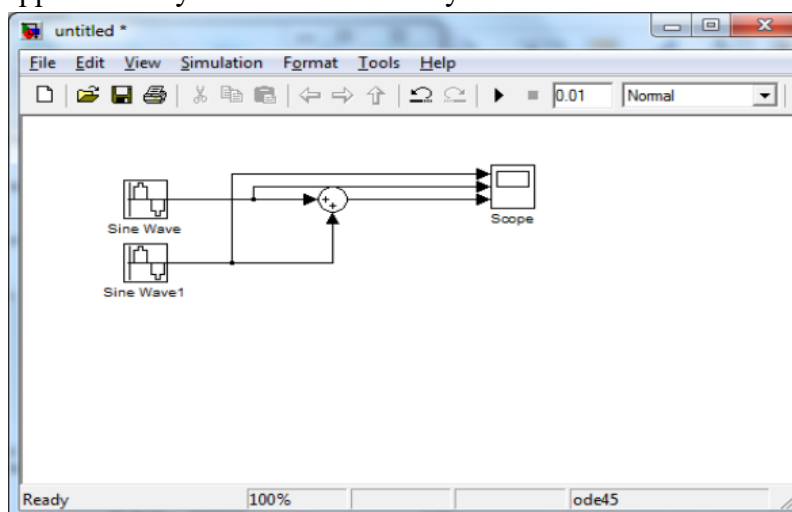


Figure 13: Summation of Two Sine Waves

EXPERIMENT NO. 1

Generation of Basic Signals

Aim: To generate Sine, Square, Triangular and Ramp signals.

Theory: In communication systems, several basic signals are commonly used for various purposes. These signals are used for testing, synchronization, modulation, and information transport. Among the fundamental signals few are described.

Sine Wave: Sine waves are continuous, periodic signals characterized by their smooth, oscillatory waveform. They are fundamental in analog communication systems, serving as carriers for transmitting information through modulation techniques such as Amplitude Modulation (AM), Frequency Modulation (FM), and Phase Modulation (PM).

Square Wave: Square waves feature sharp changes in direction between the high and low discrete levels. They are frequently utilized in digital communication systems and are composed of odd harmonics. Square waves are employed as clock signals for synchronizing the transmission and reception of digital data. Additionally, they are employed in encoding processes using pulse width modulation (PWM) in digital-to-analog conversion.

Triangle Wave: Triangle waves are characterized by their linearly increasing and decreasing slopes, which combine to form a triangle. They are used in frequency modulation synthesis to create musical tones and in modulation/demodulation processes, particularly in phase-locked loop (PLL) circuits for frequency tracking and synchronization.

Ramp Signal: Ramp signals exhibit a linearly increasing or decreasing amplitude over time. They are used in communication systems for pulse shaping, where they shape transmitted pulses to meet specific bandwidth requirements and minimize distortion. Radar systems also use ramp signals to generate linear frequency modulation (LFM) waveforms for estimating target velocity and range.

These basic signals serve as building blocks for designing and implementing communication systems, enabling the transmission, reception, processing, and analysis of information across various media and channels. Each signal type has specific characteristics that make it suitable for particular applications within communication systems.

Simulink Model for basic signals generation:

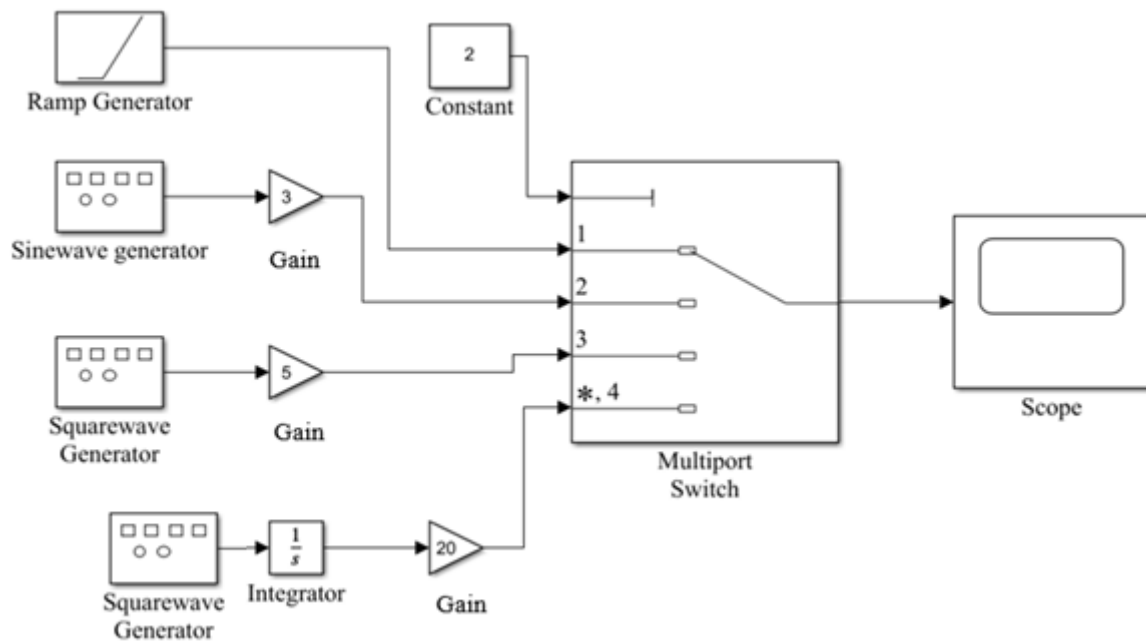


Figure 1 a) Simulink model for Signal Generation

Procedure:

1. Aim is to generate basic signals used in communication systems.
2. Using Library browser signal generators are chosen. Sinewave, square wave and Ramp generators are available in Simulink library.
3. To generate triangular wave, a sinewave is generated and is integrated with the help of an Integrator block.
4. Generated signals are routed to output through a multipoint switch/ Multiplexer. Based on the channel selected in Mux the signal gets displayed in the oscilloscope.
5. Figure 1 a) depicts the Simulink model of the experiment.
6. Figure 1 b), 1 c), 1 d) and 1 e) depicts the expected Ramp, Sine, Square and Rectangular waveforms respectively.

Expected Waveforms:

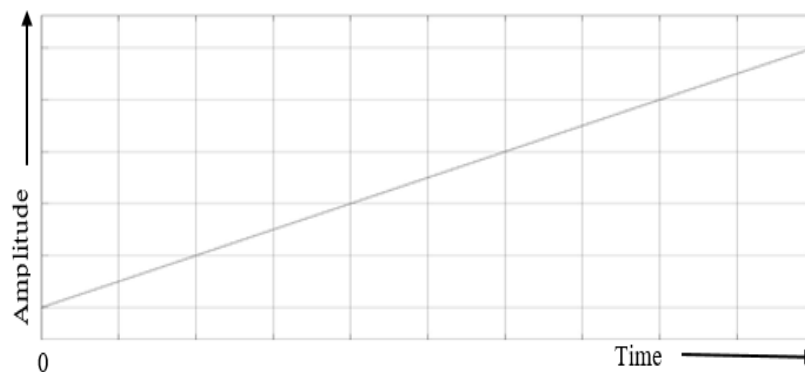


Figure 1 b) Expected Ramp Output

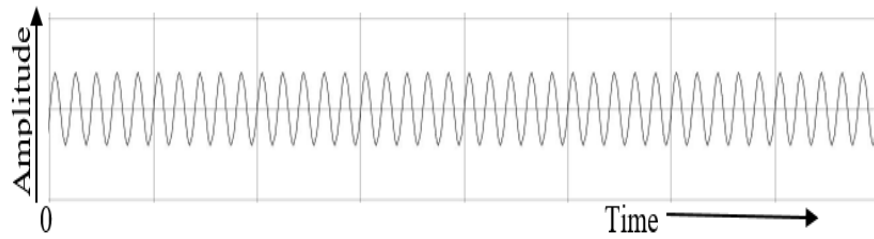


Figure 1 c) Expected Sine Output

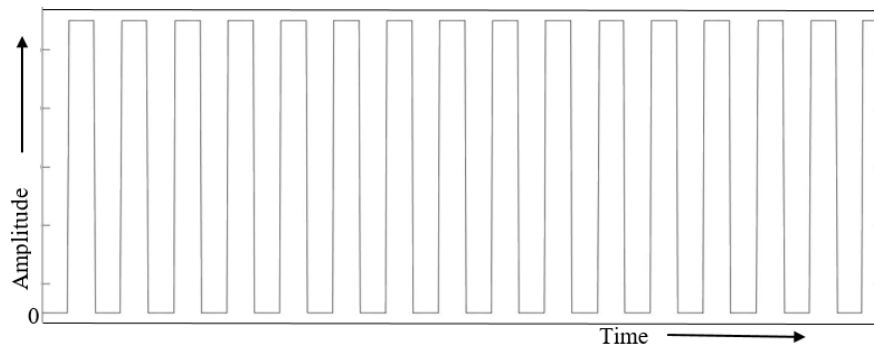


Figure 1 d) Expected Square Output

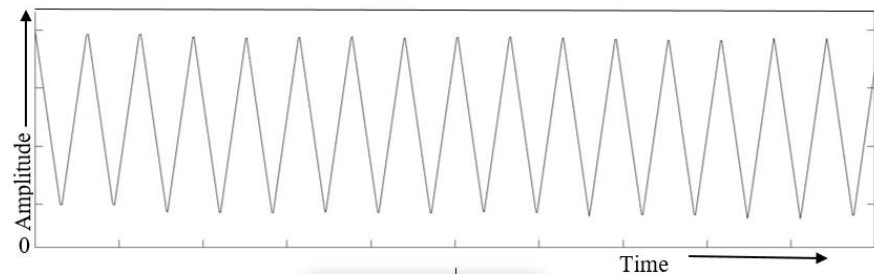


Figure 1 e) Expected Triangular Output

Result:

EXPERIMENT NO. 2

Generation and Detection of DPSK Using Simulink

Aim: To generate and detect DPSK signals using Simulink

Theory: The phase of the modulated signal is shifted in relation to the preceding signal element in differential phase shift keying, or DPSK. Here, no reference signal is considered.

DPSK Modulator: DPSK is a technique of BPSK, in which there is no reference phase signal. Here, the transmitted signal itself can be used as a reference signal. Figure 2.a) is the diagram of DPSK Modulator.

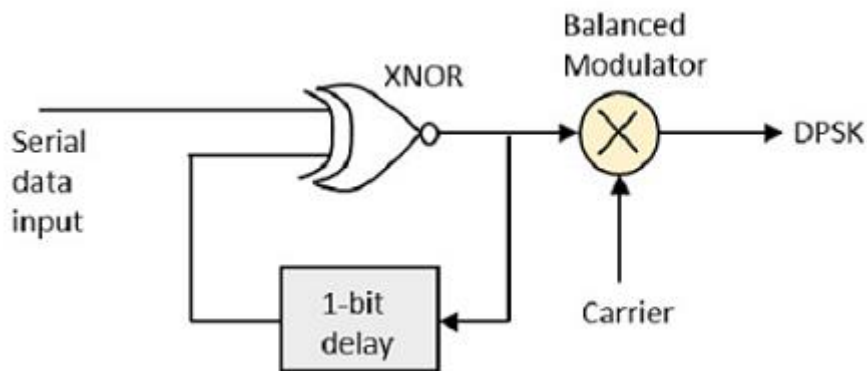


Figure 2 a) DPSK Modulator

DPSK Demodulator: In DPSK demodulator, the phase of the reversed bit is compared with the phase of the previous bit. Figure 2 b) is the block diagram of DPSK demodulator.

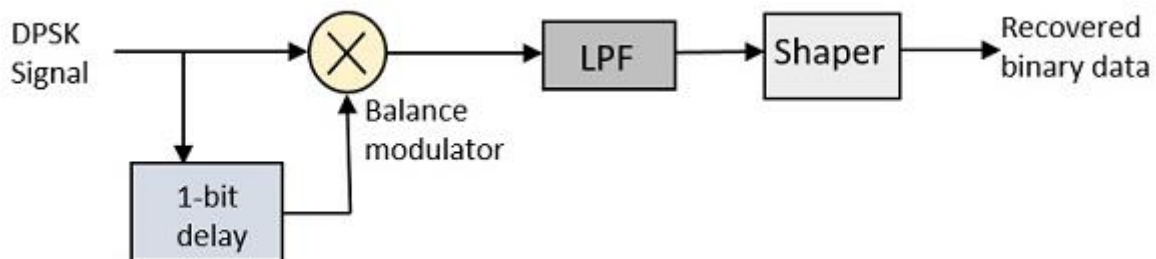


Figure 2 b) DPSK Demodulator

Figure 2 c) illustrates an example of DPSK modulation and Demodulation.

Binary Data		0	0	1	0	0	1	0	0	1	1
Differentially Encoded Data	1	0	1	1	0	1	1	0	1	1	1
Phase of DPSK	0	π	0	0	π	0	0	π	0	0	0
Shifted Differentially encoded Data d_{k-1}		1	0	1	1	0	1	1	0	1	1
Phase of shifted Data		0	π	0	0	π	0	0	π	0	0
Phase Comparision Output		-	-	+	-	-	+	-	-	+	+
Detected Binary Seq.		0	0	1	0	0	1	0	0	1	1

Figure 2 c) Example of DPSK Modulation and Demodulation

Simulink Model for DPSK Generation and Detection

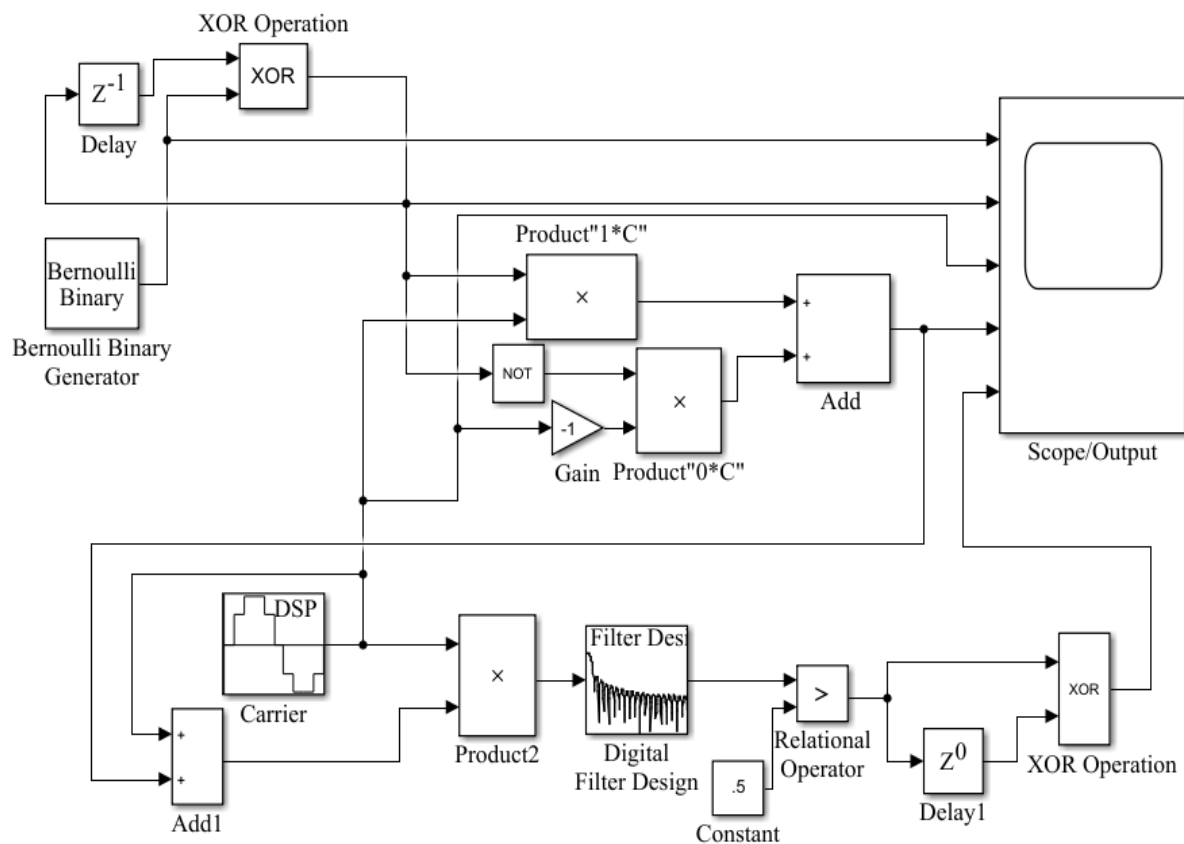


Figure 2 d) Simulink model for the Generation and Detection of DPSK

Procedure:

1. DPSK involves differential encoding and phase shift keying. Differential encoding could be done using an XOR gate or XNOR gate. Here, an XOR gate is used.
2. Bernoulli Binary Generator is used to generate random binary input.
3. Encoded input is multiplied with carrier to get ASK output for binary bits “1”. Encoded data is inverted and multiplied with an 180° phase shifted carrier to get ASK output for encoded “0” bits. Both product outputs are added to get DPSK output.
4. At the decoder end the DPSK output is added with carrier to get ASK output for received bits “1”. The resultant signal is multiplied with the carrier and filtered using an FIR filter.
5. The resultant signal is then compared with the threshold to regenerate binary output.
6. The regenerated binary output is differentially decoded to get back the original binary input.
7. Figure 2 d) depicts the Simulink model for the generation and detection of DPSK. Figure 2 e) represents the various expected input and output signals.

Output:

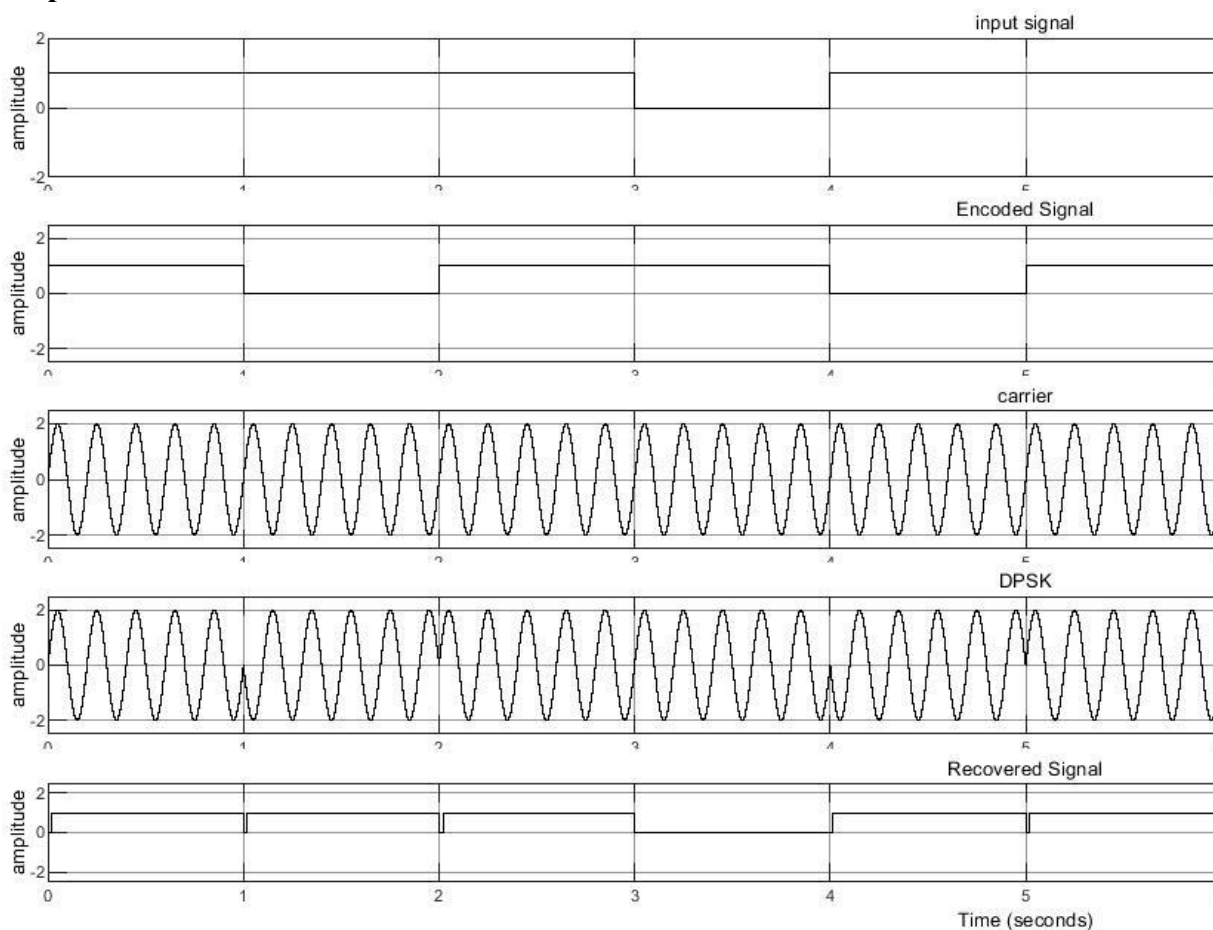


Figure 2 e) Expected Input and Output Waveforms

Result:

EXPERIMENT NO. 3

Generate a multiband signal using the Communications Toolbox

Aim: To generate a multiband signal using the Communications Toolbox.

Theory: Multi-band signals are bandlimited signals that possess an additional structure in the frequency domain. The spectral support of a multi-band signal is restricted to several continuous intervals. Each of these intervals is called a band and it is assumed that no information resides outside the bands.

The explosive growth of consumer demand for higher data rates in mobile applications leads to higher transmission rates. Most modern wireless standards include a technique to enhance the data capacity by combining two or more carriers into one data channel. This technique is called carrier aggregation in 5G and LTE terminology, and channel bonding in Wi-Fi® terminology.

This figure illustrates three different types of carrier aggregation.

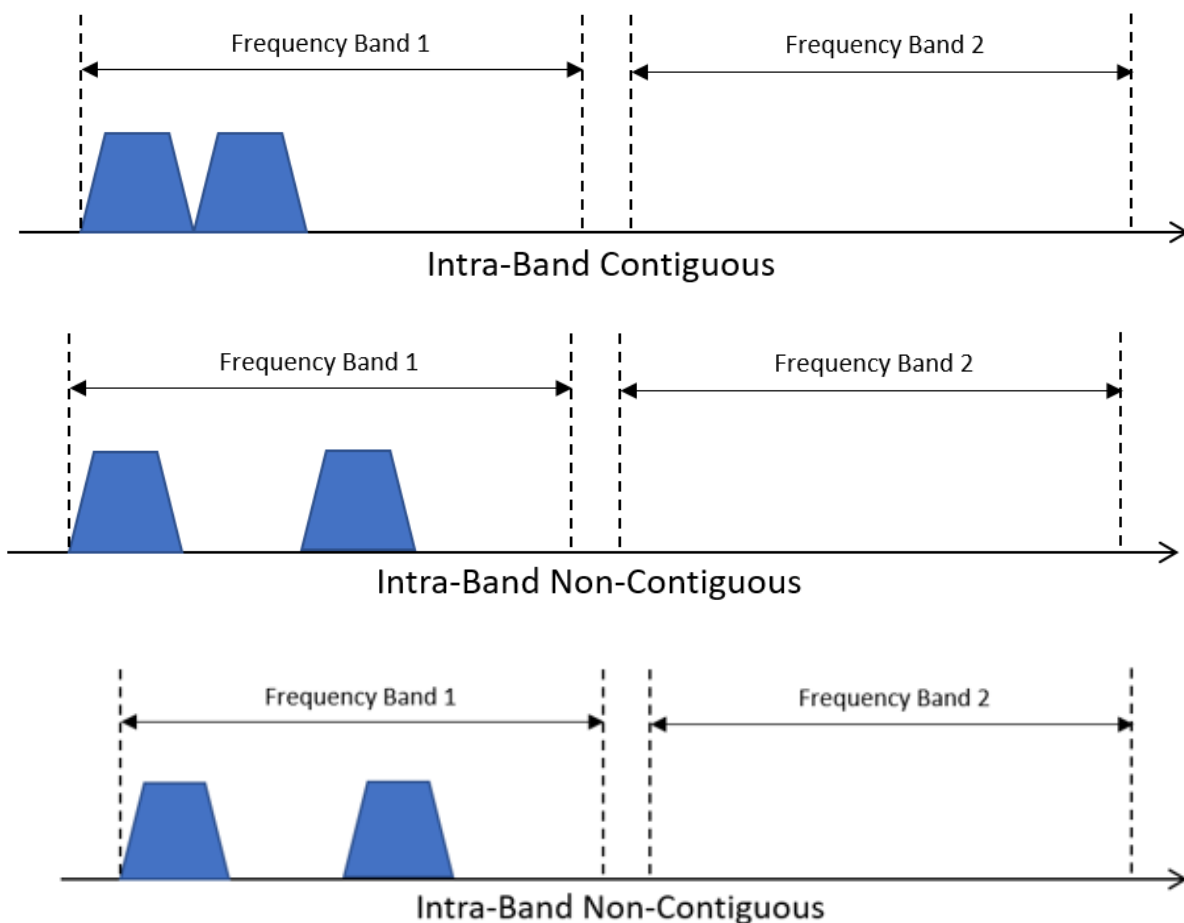
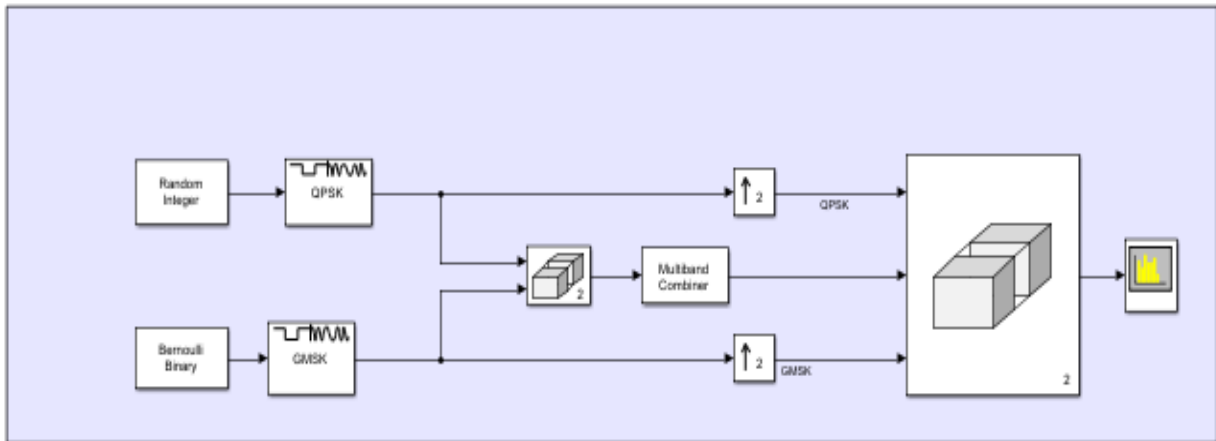
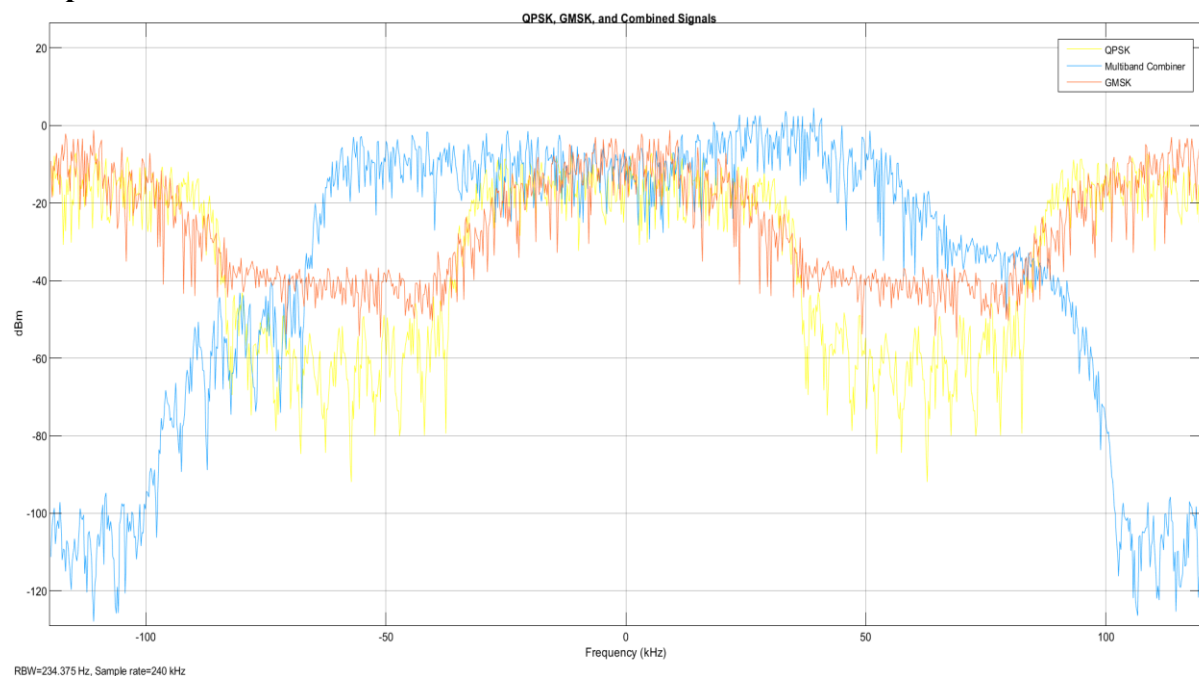


Fig. 3 a) Different types of carrier aggregation

Simulink Model to generate multiband signal:**Fig. 3 b) Simulink model to generate multiband signal****Procedure:**

1. Connect the blocks as shown in the diagram.
2. In random generator, use set size of 4.
3. Set phase offset of $\pi/4$ for QPSK modulator. Choose input type as integer.
4. In the GMSK block, input type must be bit, phase offset is zero, pulse length of 4, samples per symbol is 2.
5. Observe the output in the spectrum Analyzer.

Output:**Fig. 3 c) multiband signal****Result:**

EXPERIMENT NO. 4

Random noise generation using Simulink & display histogram plots of Gaussian and Uniform noise

Aim: To generate Random noise using Simulink & display histogram plots of Gaussian and Uniform noise.

Theory: You can generate noise for communication system modeling using the MATLAB® Function block with a random number generator. This example generates and displays histogram plots of Gaussian, Rayleigh, Rician, and Uniform noise.

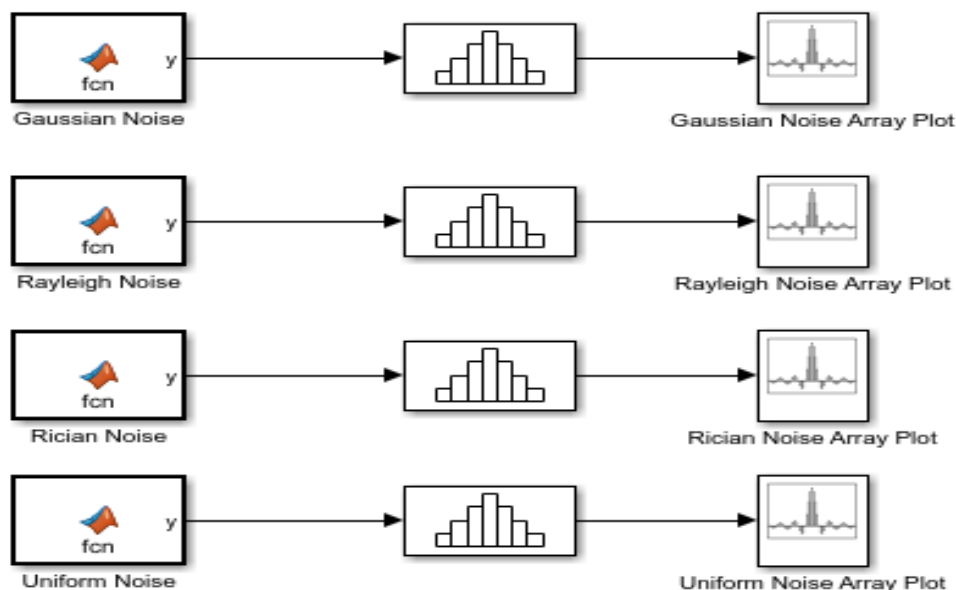


Fig. 4 a) Random Noise generation using Simulink

For each MATLAB Function block the **Samples per frame** parameter maps to its underlying function argument *spf*. Similarly, **Seed** maps to *seed*.

The **Gaussian Noise** MATLAB Function block maps the **Power (dBW)** parameter to *p*, and defines the function

$$y = wgn(spf, 1, p)$$

The **Rayleigh Noise** MATLAB Function block maps the **Sigma** parameter to *alpha*, and defines the function

$$y = \text{abs}(\alpha * (\text{randn}(spf, 1) + 1i * \text{randn}(spf, 1)))$$

The **Rician Noise** MATLAB Function block maps the **Rician K-factor** parameter to *K* and the **Sigma** parameter to *s*, and defines the function

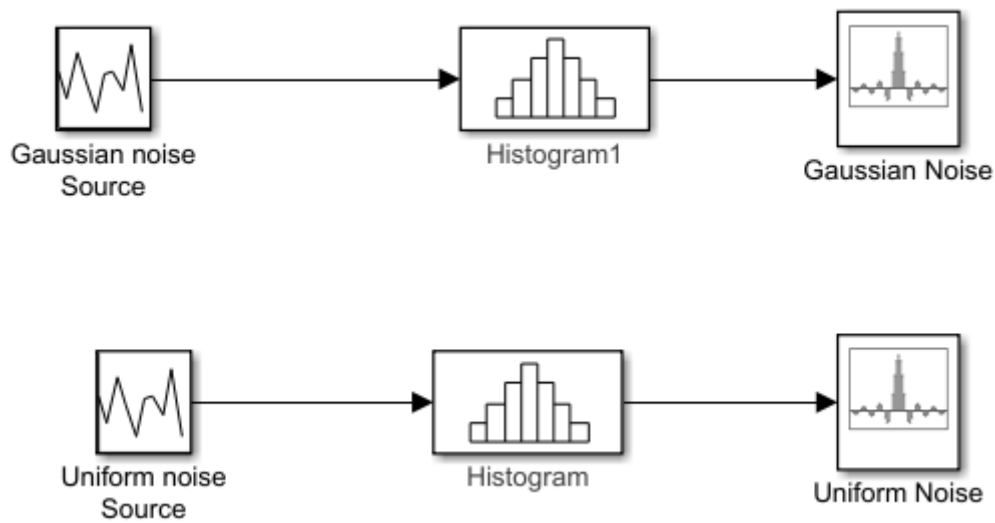
$$m1 = \text{sqrt}(2 * K) * s$$

$$m2 = 0$$

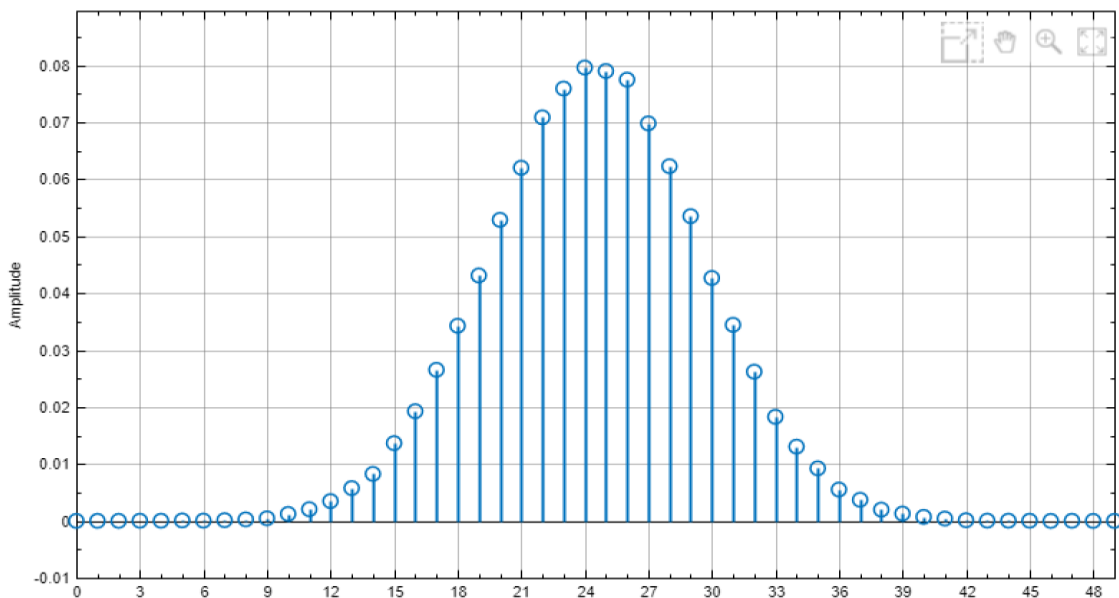
$$y = \text{sqrt}((s^2 * \text{randn}(spf, 1) + m1)^2 + (s^2 * \text{randn}(spf, 1) + m2)^2)$$

$$y = lb + (ub - lb) * \text{rand}(spf, 1)$$

The model generates histogram plots to show the noise distribution across the spectrum for each noise generator.

Simulink Model to generate multiband signal:**Fig. 4 b) Random Noise generation using Simulink****Procedure:**

1. Connect the blocks as shown in the diagram.
2. For Gaussian noise set the source type as Gaussian and mean of 0 and variance of 1.
3. For Uniform noise set the source type as Uniform and mean of 0 and variance of 1.
4. Set the lower and upper limit of histogram to -5 and 5 respectively and number of bins as 50.
5. Set find the histogram over as each column.
6. Observe the output in the Array plot.

Output:**Fig. 4 c) Histogram of Gaussian Noise using Simulink**

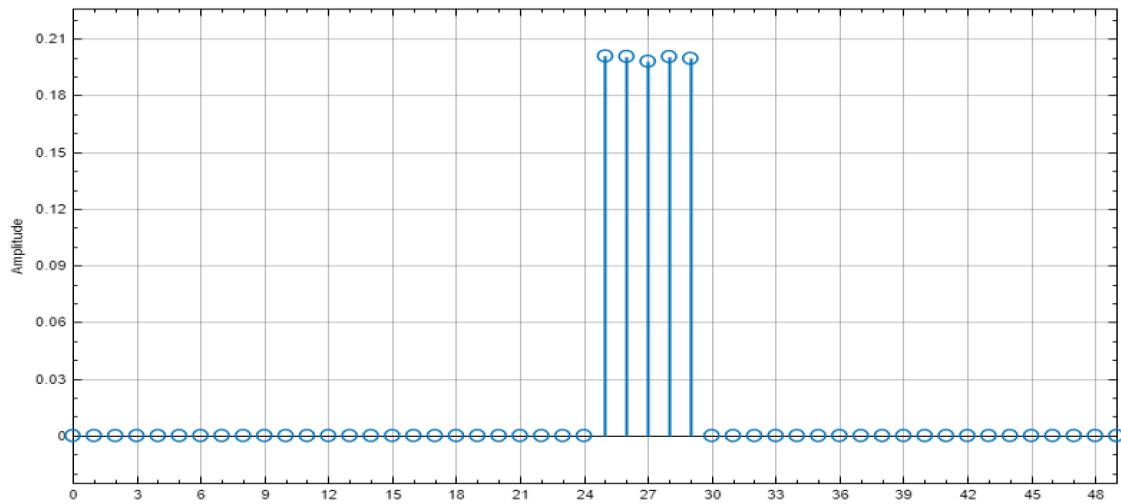


Fig. 4 d) Histogram of Uniform Noise using Simulink

Result:

EXPERIMENT NO. 5

Perform Equalization, Convolution and Cyclic Prefix addition on basic OFDM

Aim: To perform equalization, convolution and cyclic prefix addition on basic OFDM.

Theory: Equalization is the reversal of distortion incurred by a signal transmitted through a channel. Equalizers are used to render the frequency response flat from end-to-end. When a channel has been equalized the frequency domain attributes of the signal at the input are faithfully reproduced at the output. Telephones, DSL lines and television cables use equalizers to prepare data signals for transmission. Equalizers are critical to the successful operation of electronic systems such as analog broadcast television. In this application the actual waveform of the transmitted signal must be preserved, not just its frequency content. Equalizing filters must cancel out any group delay and phase delay between different frequency components.

Convolution is a mathematical way of combining two signals to form a third signal. It is the single most important technique in Digital Signal Processing. Using the strategy of impulse decomposition, systems are described by a signal called the impulse response. Convolution is important because it relates the three signals of interest: the input signal, the output signal, and the impulse response. Convolution provides the mathematical framework for DSP.

Cyclic prefix (CP) is a technique used in orthogonal frequency division multiplexing (OFDM) systems to mitigate the effects of multipath fading and inter-symbol interference (ISI). OFDM is a modulation scheme that divides the available bandwidth into multiple subcarriers, each carrying a fraction of the data. OFDM can achieve high spectral efficiency and robustness against frequency-selective fading, but it also introduces some challenges, such as high peak-to-average power ratio (PAPR) and sensitivity to frequency and timing offsets. CP is a portion of the end of an OFDM symbol that is copied and appended to the beginning of the same symbol. The length of the CP is usually chosen to be equal or longer than the maximum expected delay spread of the channel, which is the difference between the arrival times of the earliest and the latest multipath components. By adding CP, the OFDM symbol duration is extended, and the cyclic property of the discrete Fourier transform (DFT) is preserved.

CP works by creating a circular convolution between the transmitted OFDM symbol and the channel impulse response. This means that the received OFDM symbol can be expressed as a linear convolution of the original OFDM symbol and the channel impulse response, plus some additive noise. This simplifies the equalization process at the receiver, since a simple one-tap frequency-domain equalizer can be applied to each subcarrier, instead of a complex time-domain equalizer. CP also eliminates ISI, as long as the CP length is longer than the delay spread, because the overlapping parts of the adjacent symbols are identical and cancel out.

CP offers several advantages in OFDM systems, such as reducing complexity and cost of the receiver, improving performance and reliability in frequency-selective fading channels, enabling simple and efficient channel estimation techniques, and allowing for flexible and adaptive allocation of subcarriers, power, and modulation schemes. Specifically, CP only requires a one-tap equalizer per subcarrier instead of a multi-tap equalizer for the whole symbol. It also preserves the orthogonality of the subcarriers and eliminates ISI.

Furthermore, the channel response can be easily obtained from the frequency-domain equalizer coefficients. Lastly, each subcarrier can be independently modulated and equalized according to the channel conditions. CP can also be disadvantageous for OFDM systems, as it reduces the spectral efficiency and data rate. This is due to the overhead and redundancy it introduces into the transmitted signal, which is dependent on the ratio of CP length to OFDM symbol length. Furthermore, CP increases the PAPR of OFDM systems, leading to nonlinear distortion and power inefficiency in the transmitter. Lastly, precise synchronization and alignment of the CP boundaries is necessary at the receiver to avoid ICI and maintain subcarrier orthogonality. Synchronization errors can arise from frequency and timing offsets, phase noise, or Doppler shifts.

MATLAB program:

```
clc;
clear;
close all;
u1 = 1:8;
h = [0.4 1 0.4];
figure
subplot(2,1,1)
stem(u1);
axis([0 10 0 10])
title("Input signal")
subplot(2,1,2)
stem(h,'^');
axis([0 10 0 2])
title("Channel impulse response")
N = length(u1);
y11 = conv(u1,h);
yc1 = cconv(u1,h,N);
figure;
stem(y11,"x")
hold on;
stem(yc1,"o")
title(["Convolution Results, N=",int2str(N)])
legend ("Linear","Circular","Location","northwest")
L = length(h);    % Length of channel
N = length(u1);   % Length of input signal
```

```

ucp = u1(N-L+1:N); % Use last samples of input signal as the CP
u2 = [ucp u1];    % Prepend the CP to the input signal
yl2 = conv(u2,h); % Convolution of input+CP and channel
yl2 = yl2(L+1:end); % Remove CP to compare signals
figure;
stem(yl2,"x")
hold on;
stem(yc1,"o")
title("Convolution Results with Cyclic Prefix")
legend("Linear","Circular","Location","northwest")
if max(yc1 - yl2(1:N)) < 1e-8
    disp("Linear and circular convolution sequences match.")
else
    disp("Received symbols do not match transmitted symbols.")
end

```

Output:

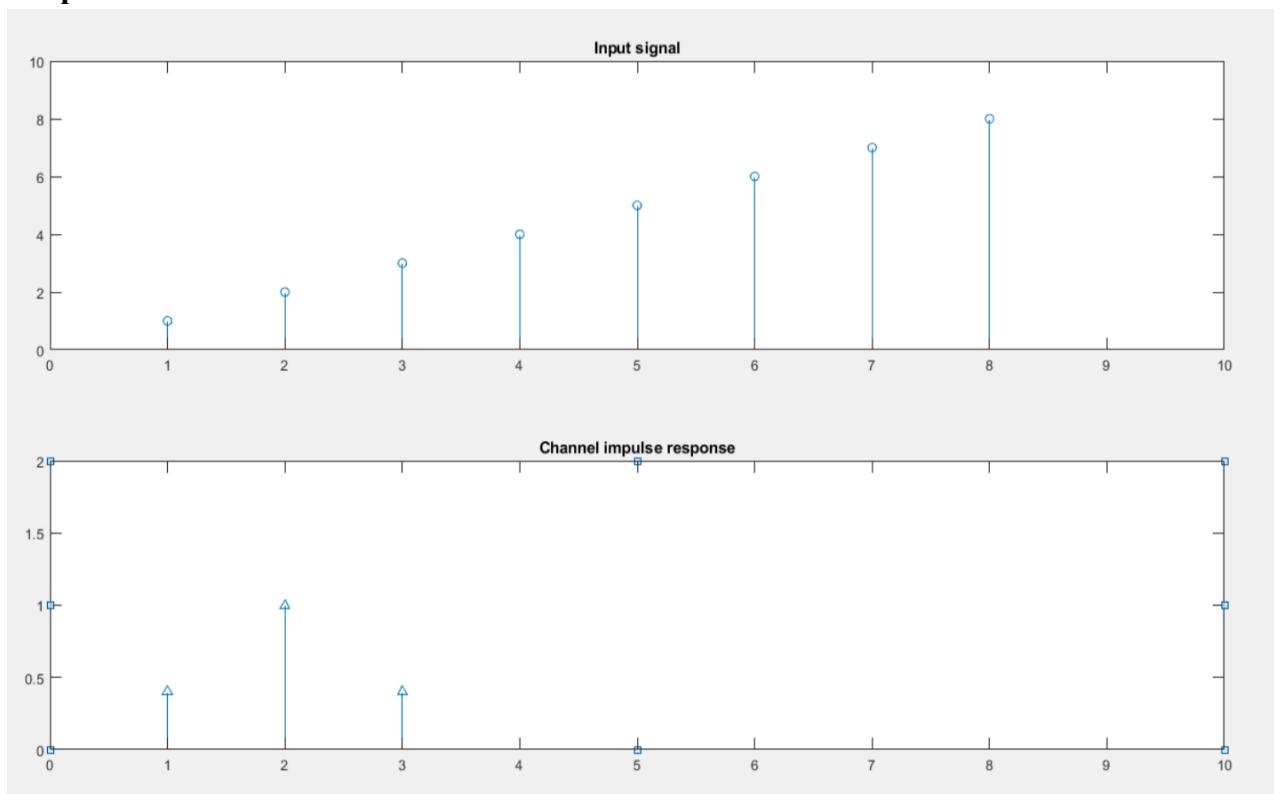


Fig. 5 a) Input and Channel Impulse Response

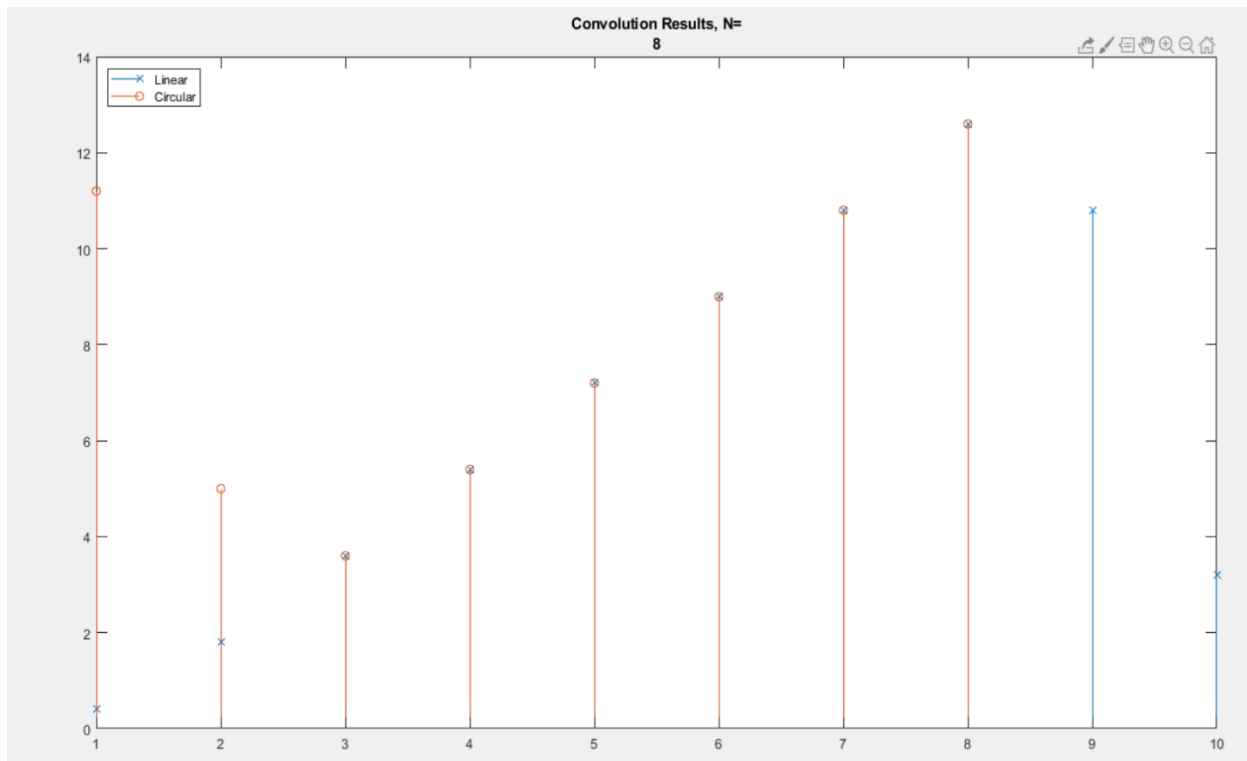


Fig. 5 b) Convolution Results

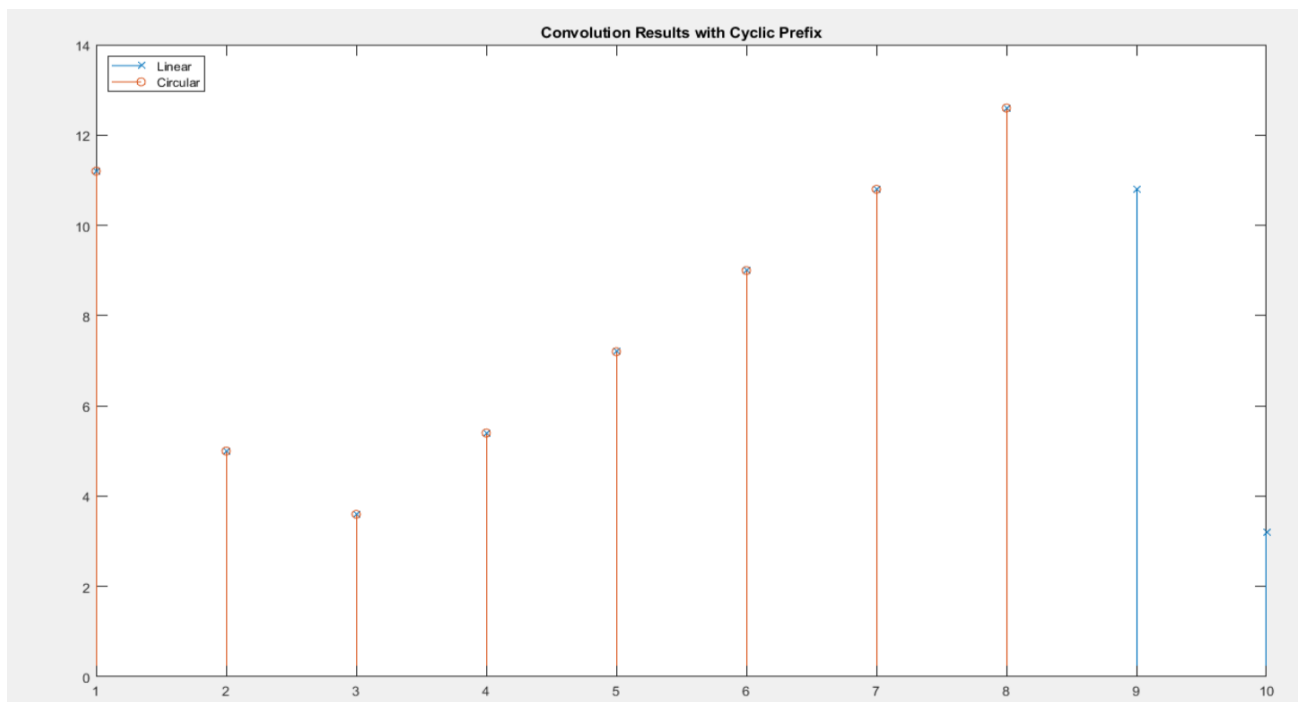


Fig. 5 c) Convolution Results with Cyclic Prefix

Result:

EXPERIMENT NO. 6

QPSK Transmitter and Receiver in Simulink

Aim: To design a QPSK Transmitter and Receiver by creating a Model in SIMULINK.

Theory: The Quadrature Phase Shift Keying is a variation of BPSK, and it is also a Double Side Band Suppressed Carrier DSBSC modulation scheme, which sends two bits of digital information at a time, called as dibits. Instead of the conversion of digital bits into a series of digital stream, it converts them into bit pairs. This decreases the data bit rate to half, which allows space for the other users.

The QPSK Modulator uses a bit-splitter, two multipliers with local oscillator, a 2-bit serial to parallel converter, and a summer circuit. Following is the block diagram for the same.

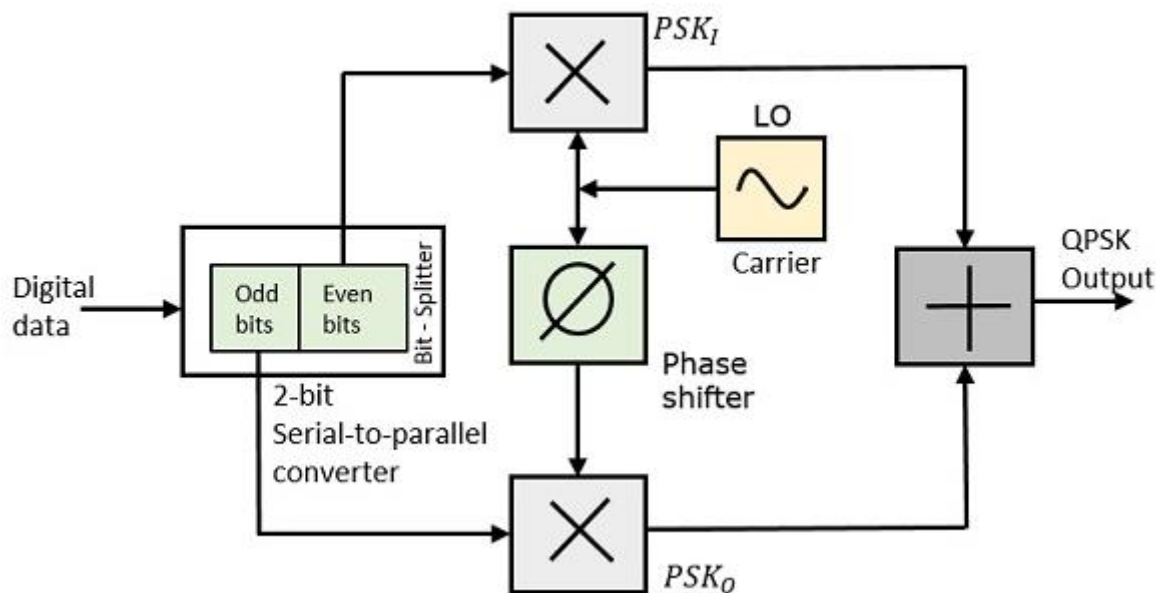


Fig. 6 a) QPSK Modulator

At the Modulator's input, the message signal's even bits (i.e., 2nd bit, 4th bit, 6th bit, etc.) and odd bits (i.e., 1st bit, 3rd bit, 5th bit, etc.) are separated by the bits splitter and are multiplied with the same carrier to generate odd BPSK (called as **PSK_I**) and even BPSK (called as **PSK_Q**). The **PSK_Q** signal is anyhow phase shifted by 90° before being modulated. The QPSK waveform for two-bits input is as follows, which shows the modulated result for different instances of binary inputs.

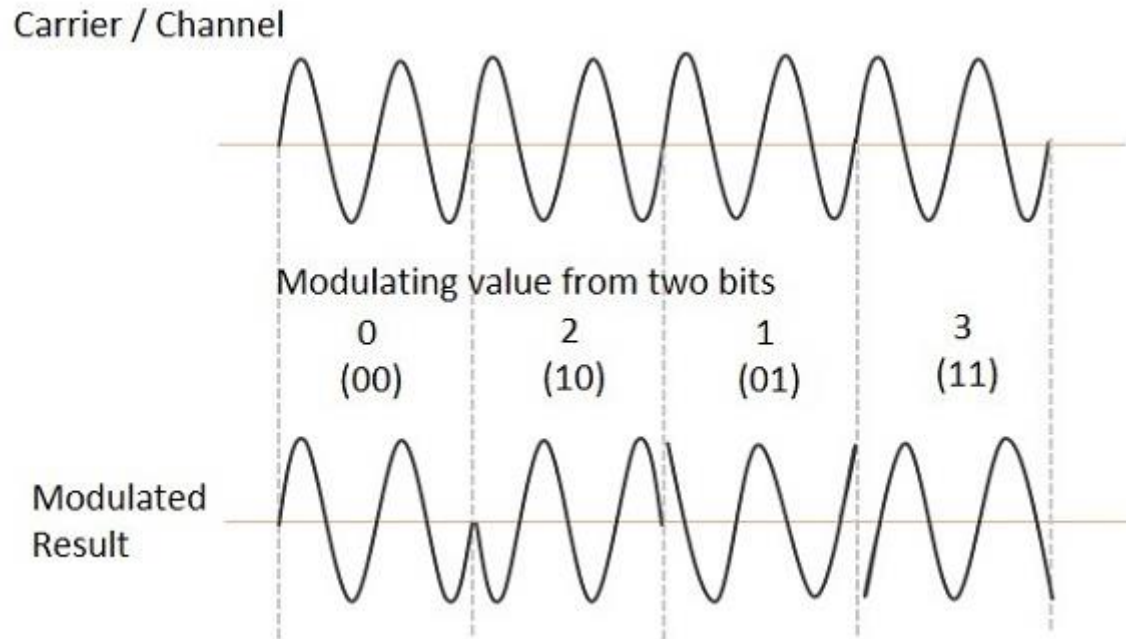


Fig. 6 b) QPSK Modulated Wave

The QPSK Demodulator uses two product demodulator circuits with local oscillator, two band pass filters, two integrator circuits, and a 2-bit parallel to serial converter. Following is the diagram for the same.

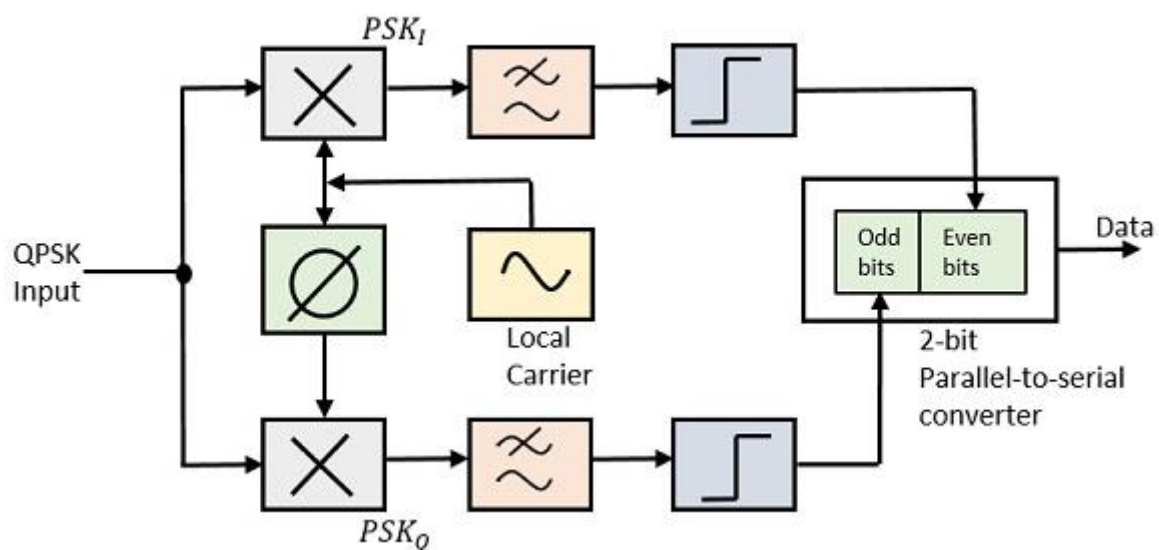


Fig. 6 c) QPSK Demodulator

The two product detectors at the input of demodulator simultaneously demodulate the two BPSK signals. The pair of bits are recovered here from the original data. These signals after processing, are passed to the parallel to serial converter.

Simulink Model for QPSK Transmitter and Receiver:

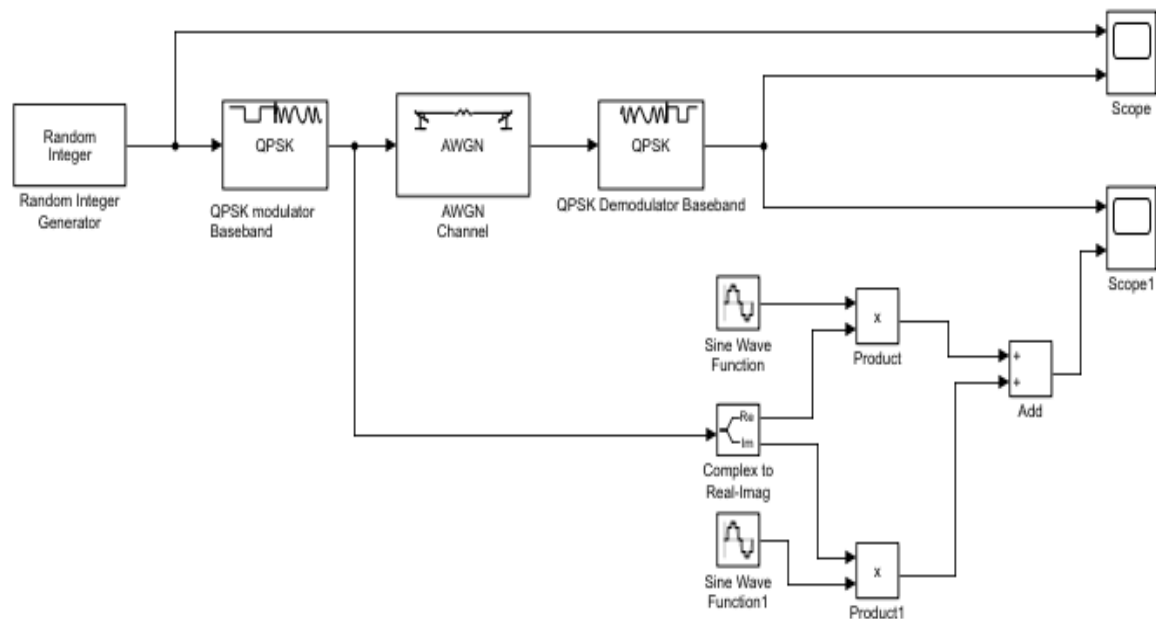


Fig. 6 d) Simulink Model for QPSK Transmitter and Receiver

Output:

Scope1:

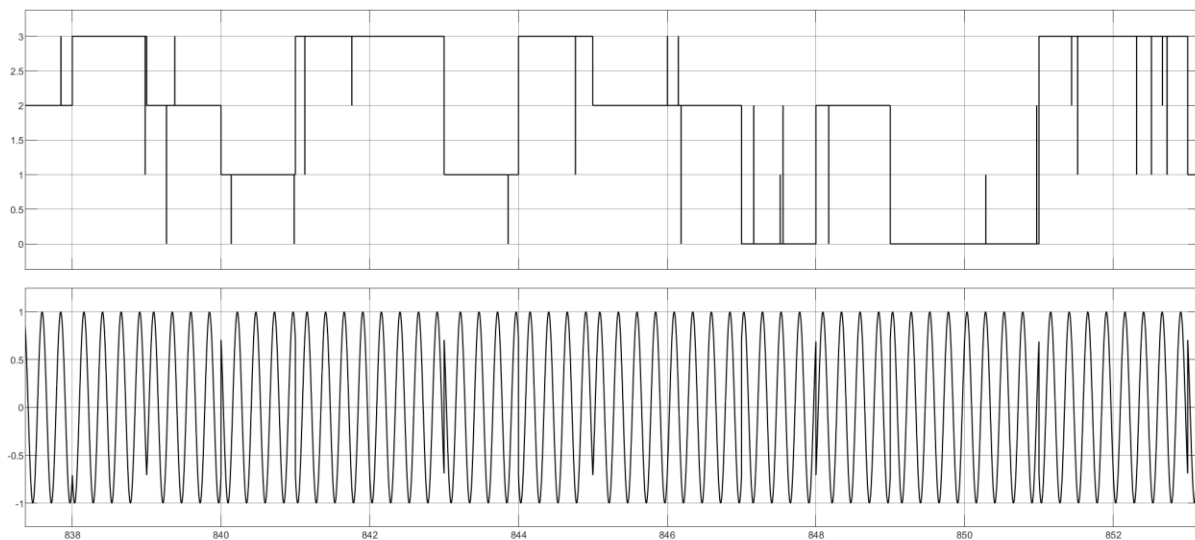


Fig. 6 e) QPSK Modulator output

Scope:

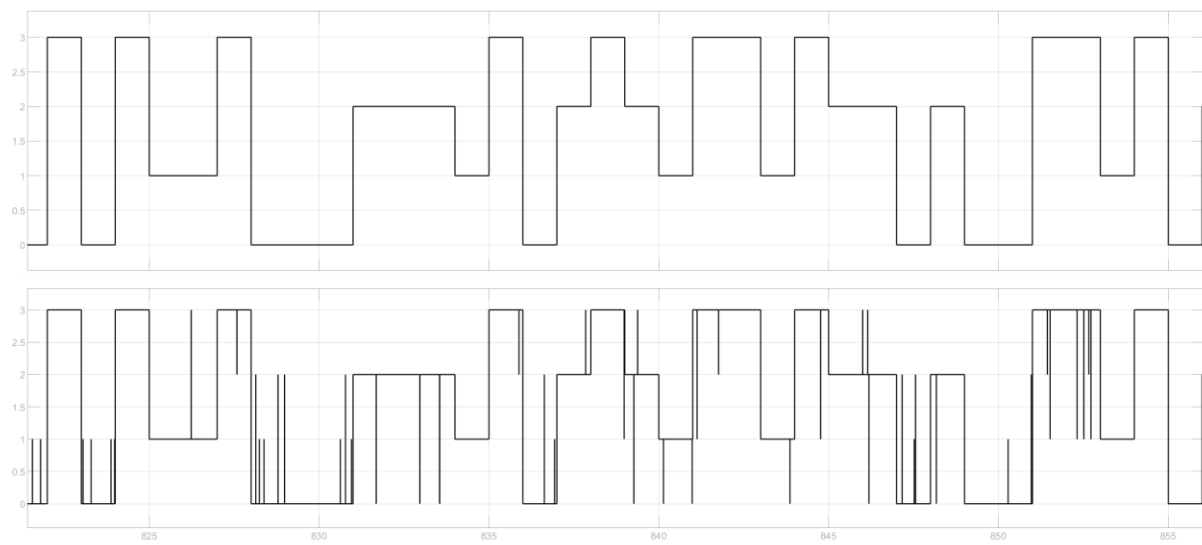


Fig. 6 f) QPSK Demodulator output

Result:

EXPERIMENT NO. 7

Design and Implementation of Low pass and High pass filter

Aim: To design and implement Low pass and High pass filter.

Theory: Low pass filter is the type of frequency domain filter that is used for smoothing the image. It attenuates the high-frequency components and preserves the low-frequency components. High pass filter is the type of frequency domain filter that is used for sharpening the image. It attenuates the low-frequency components and preserves the high-frequency components. Low-pass and high-pass filters find applications in a variety of fields including audio processing, image processing, communication systems, and biomedical signal processing. Understanding the characteristics of these filters and their applications is essential for signal-processing engineers. Figure 7 a) and 7 b) illustrate LPF and HPF.

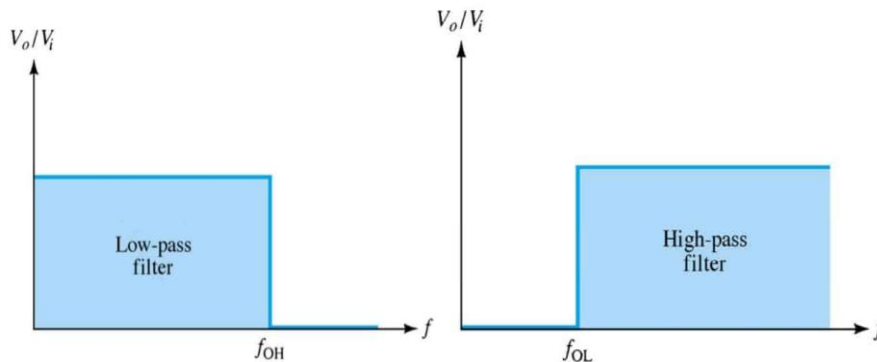


Fig.7 a) Low pass filter

Fig. 7 b) High pass filter

Simulink Model of Low pass and High pass filter:

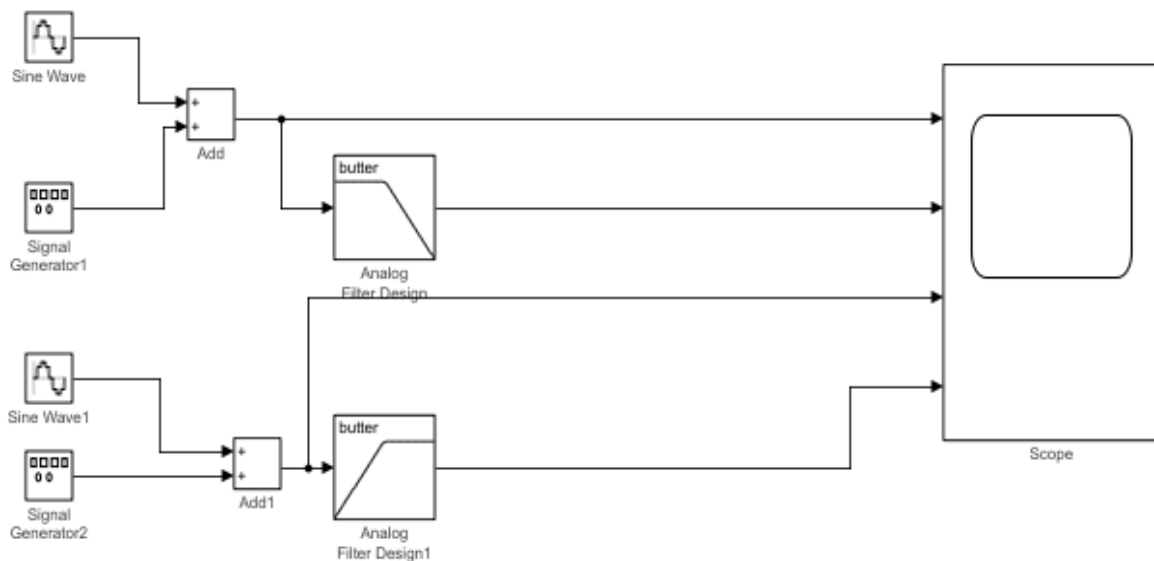


Figure 7 c) Simulink Model of LPF and HPF

Procedure:

1. Sine wave generator and random noise generator are added to produce noisy sinusoidal signal.
2. Low pass Butterworth filter of order 8, with cut-off frequency 30 rad/s is designed.
3. High pass Butterworth filter of order 8, with cut-off frequency 10 rad/s is designed.
4. The output is observed in the scope.

Output:

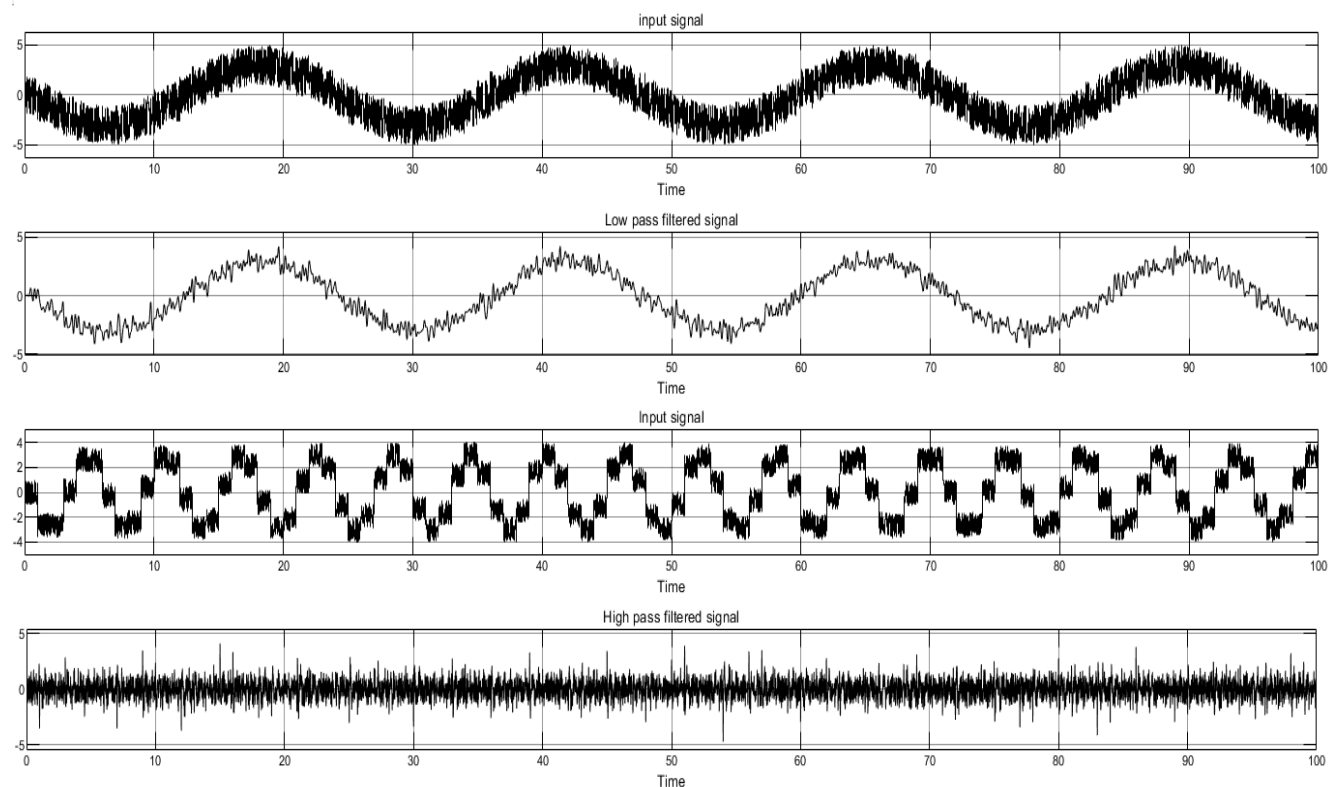


Figure 7 d) LPF and HPF outputs

Result:

EXPERIMENT NO. 8

Design and Implementation of Band Pass and Band Reject filter

Aim: To design and implement Band Pass and Band Reject filter.

Theory: A band pass filter is a combination of a high pass and an LPF. It allows only a selected range of frequencies to pass through. It is designed in such a way that the cut-off frequency of the LPF is higher than the cut-off frequency of the high-pass filter, hence allowing only a selected range of frequencies to pass through. A band stop filter also known as a notch filter is used in circuits that block only a selected range of frequencies and allows others to pass through. It would be an inverse of the band pass filter and can be created by using the same input at a high pass and an LPF. Figure 8 a) and 8 b) illustrate Band Pass Filter and Band Reject Filter.

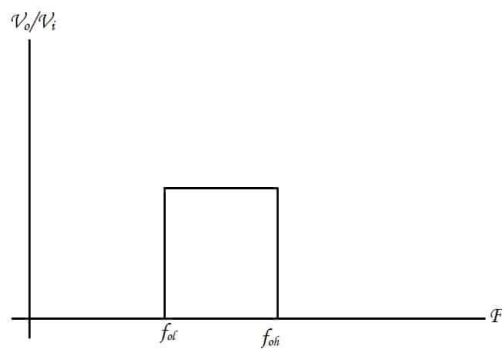


Fig.8 a) Band Pass Filter

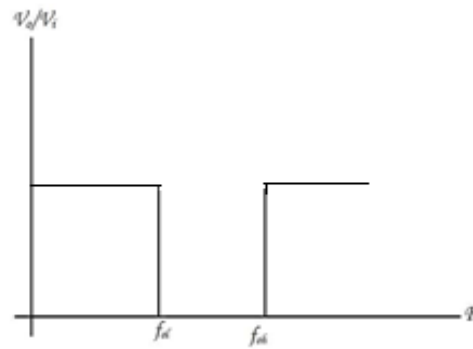


Fig. 8 b) Band Reject Filter

Simulink Model of Band Pass and Band Reject filter:

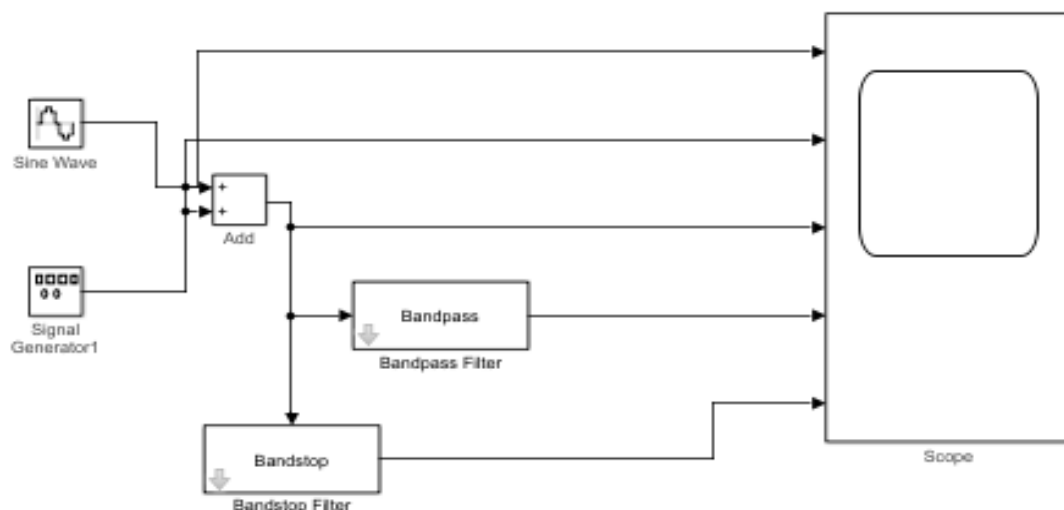


Fig. 8 c) Simulink Model of Band Pass and Band Reject filter

Procedure:

1. Sine wave generator and random noise generator are added to produce noisy sinusoidal signal.
2. To design a Band pass filter, for high pass filter, use stop band cutoff frequency of 20Hz and pass band frequency of 30Hz, and for low pass filter use pass band frequency of 40Hz and stop band frequency of 50Hz. Pass band attenuation of 1dB and stop band attenuation of 80dB is used. Use input sampling rate of 256 samples per second.
3. To design a Band stop filter, for low pass filter, use stop band cutoff frequency of 25 Hz and stop band frequency of 35Hz, for high pass use stop band frequency 40Hz and pass band frequency of 50Hz. Passband ripple is 1dB and stop band ripple is 60dB.
4. The output is observed in the scope.

Output:

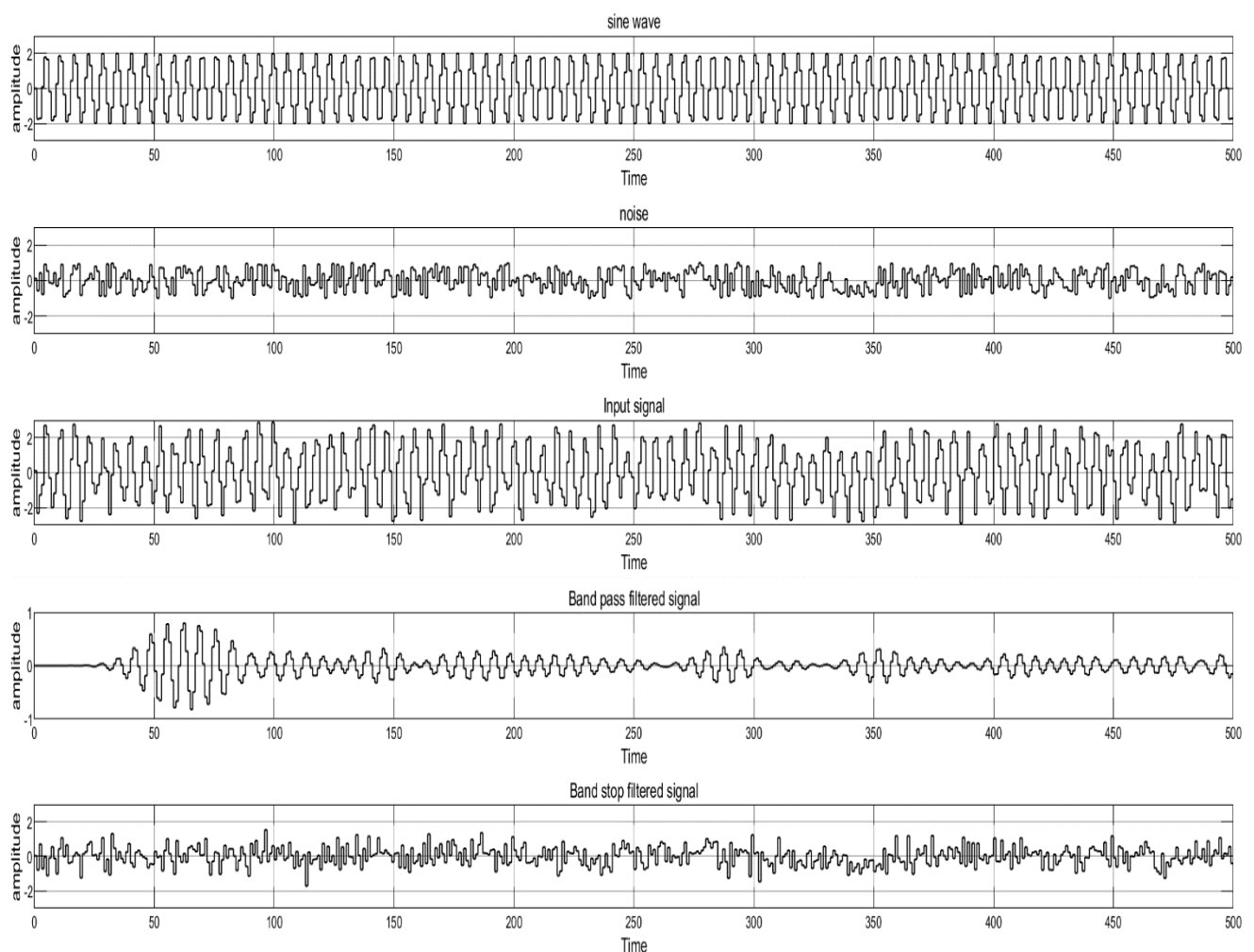


Fig. 8 d) Output Waveforms of Band Pass and Band Reject filters

Result:

EXPERIMENT NO. 9

Obtain the scatter plots & eye diagrams of a QPSK signal to visualize the signal behaviour in presence of AWGN

Aim: To obtain the scatter plots & eye diagrams of a QPSK signal to visualize the signal behaviour in presence of AWGN.

Theory: The most appropriate modulation and channel coding for a Very Small Aperture Terminal (VSAT) system is Quadrature Phase Shift Keying (QPSK). The Channel can be wire or wireless voice and data applications due to its flexible system architecture. In this paper, a Simulink based QPSK system is simulated to study the characteristic performance analysis of Additive White Gaussian Noise (AWGN) channel. Simulation study helps to visualize eye-diagram and Root Raised cosine (RRC) Filter with scatter plot. The error rate is calculated by comparing a transmitted data stream with a receive data stream with the help of delay introduction. Characteristic performance analysis is done by comparing the un-coded data with coded data using two different models. The Bit Error Rate (BER) curve for a Communication system illustrates the relationship between power in the transmitted signal in terms of signal to noise ratio (SNR) and the resulting BER for the system.

Simulink Model:

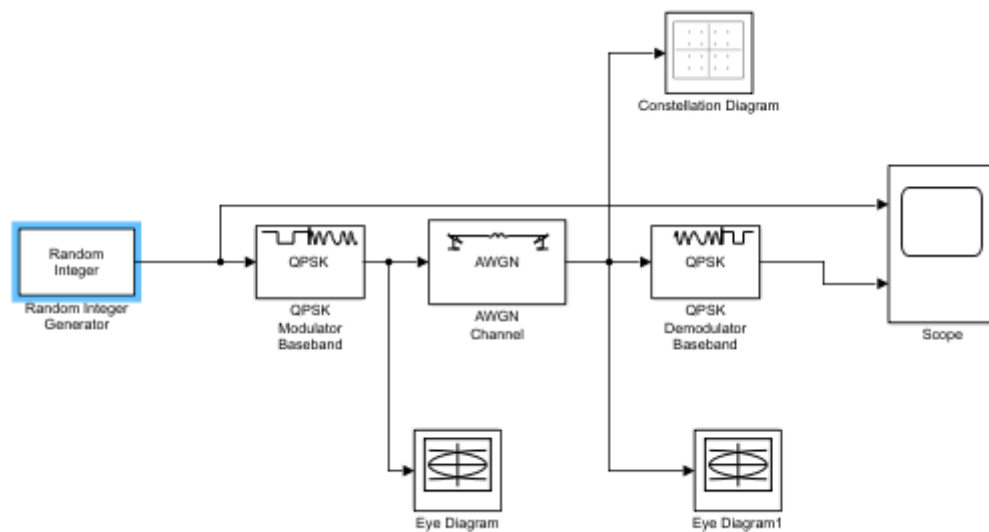


Fig. 9 a) Simulink model for QPSK Modulator and Demodulator with and without noise

Procedure:

1. Connect the blocks as shown in the block diagram.
2. For random integer, set size is 4, sample time is 0.001, samples per frame is 1.
3. For QPSK modulator and demodulator, phase offset is set as $\pi/4$, constellation ordering is set as grey and input type as integer.
4. For AWGN input seed is 67, SNR is set as 10dB.
5. Output is observed in scope.

Output:

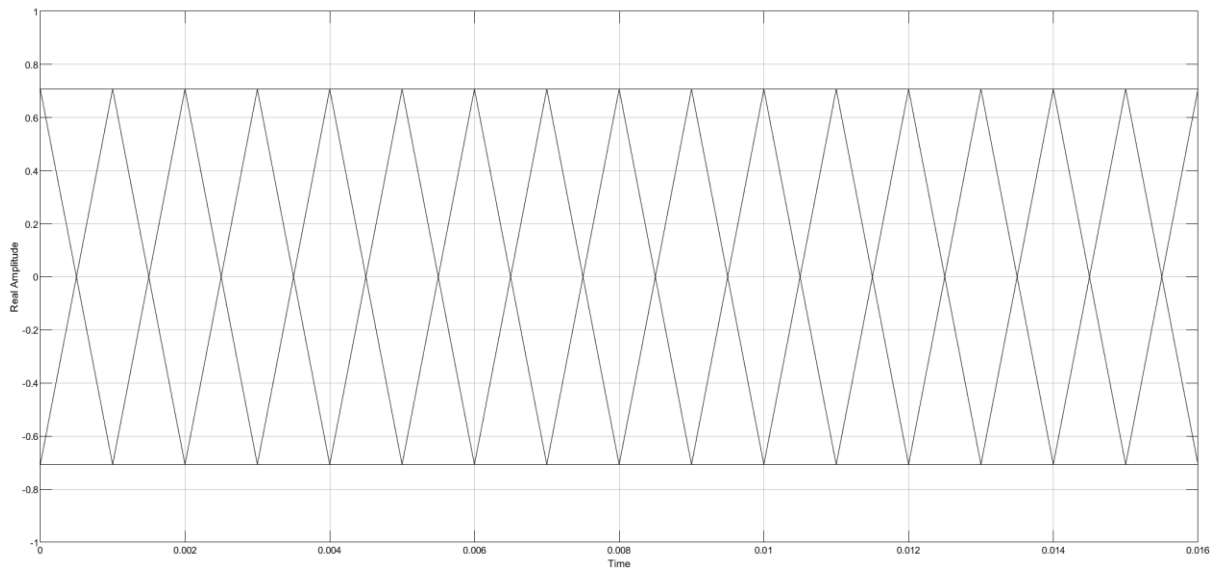


Fig. 9 b) Eye diagram of a QPSK signal without noise

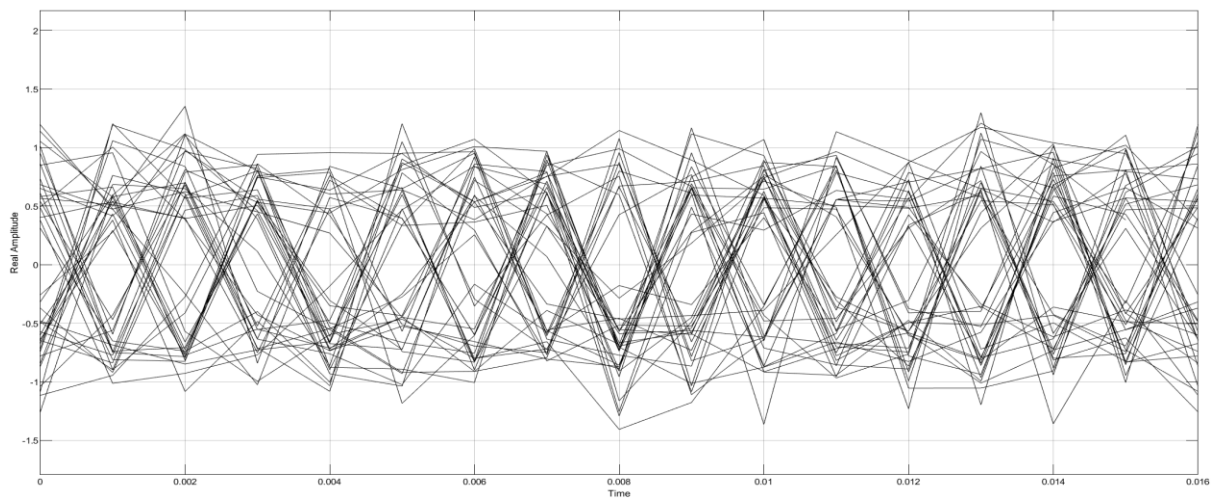


Fig. 9 c) Eye diagram of a QPSK signal with noise

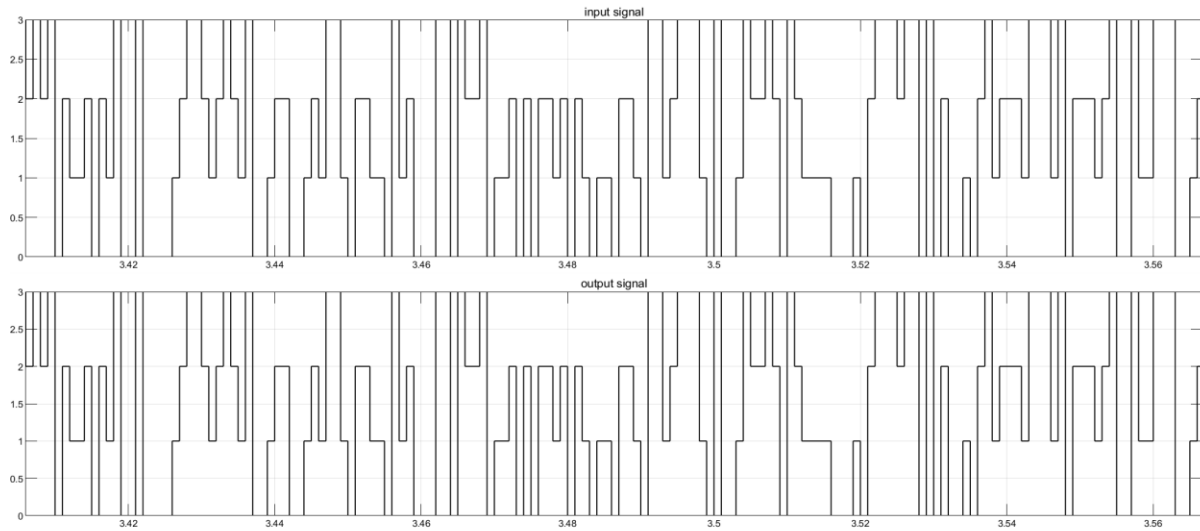


Fig. 9 d) Transmitted and Received QPSK signal with noise

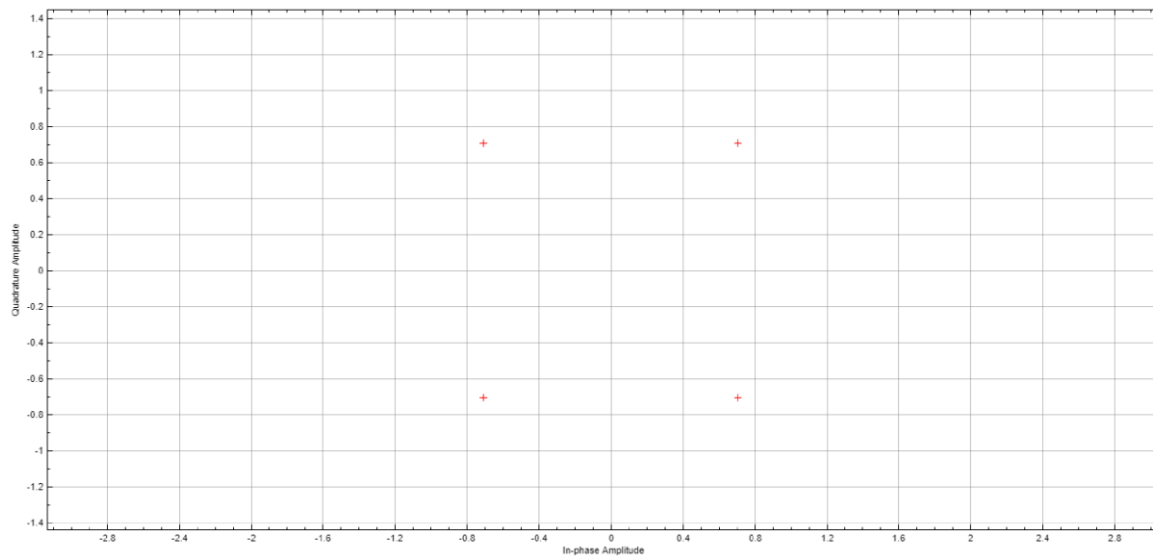


Fig. 9 e) Constellation diagram of QPSK signal with noise

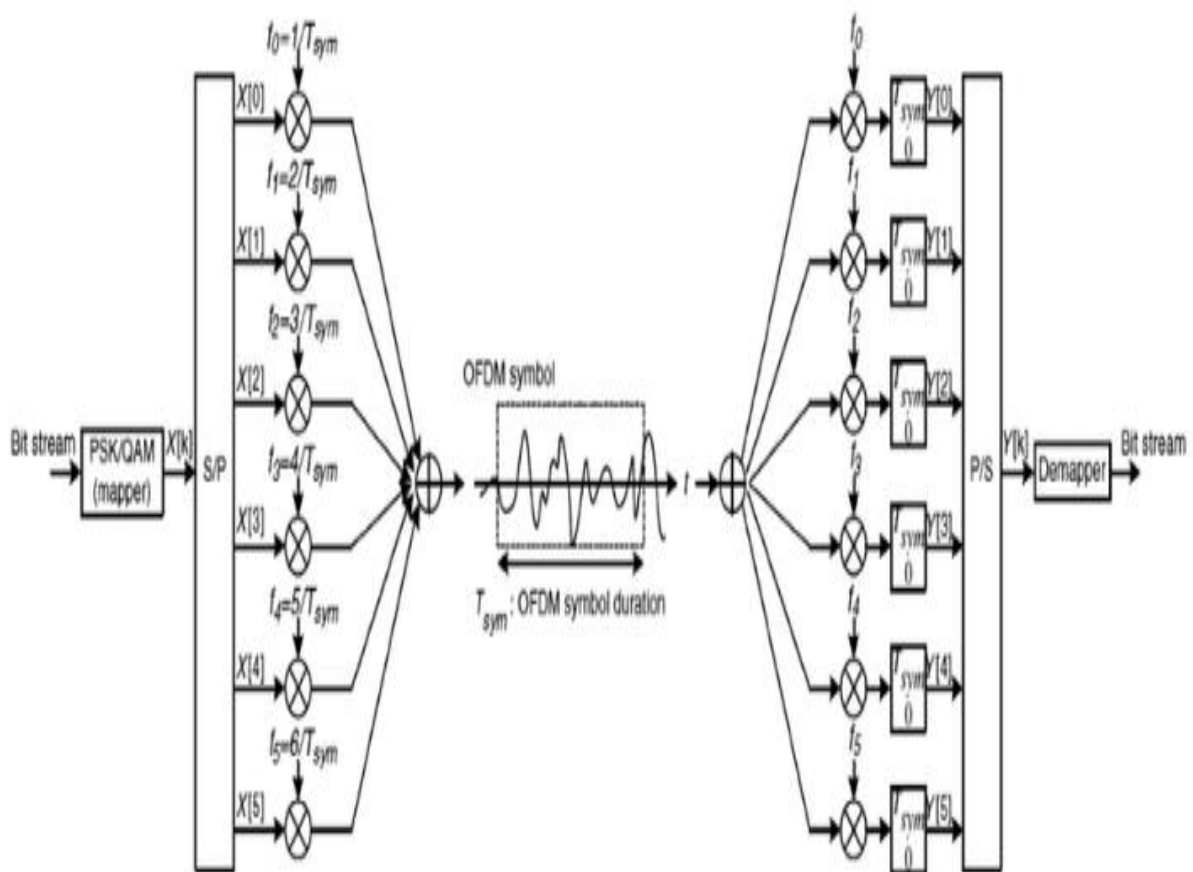
Result:

EXPERIMENT NO. 10

Perform OFDM Modulation and Demodulation

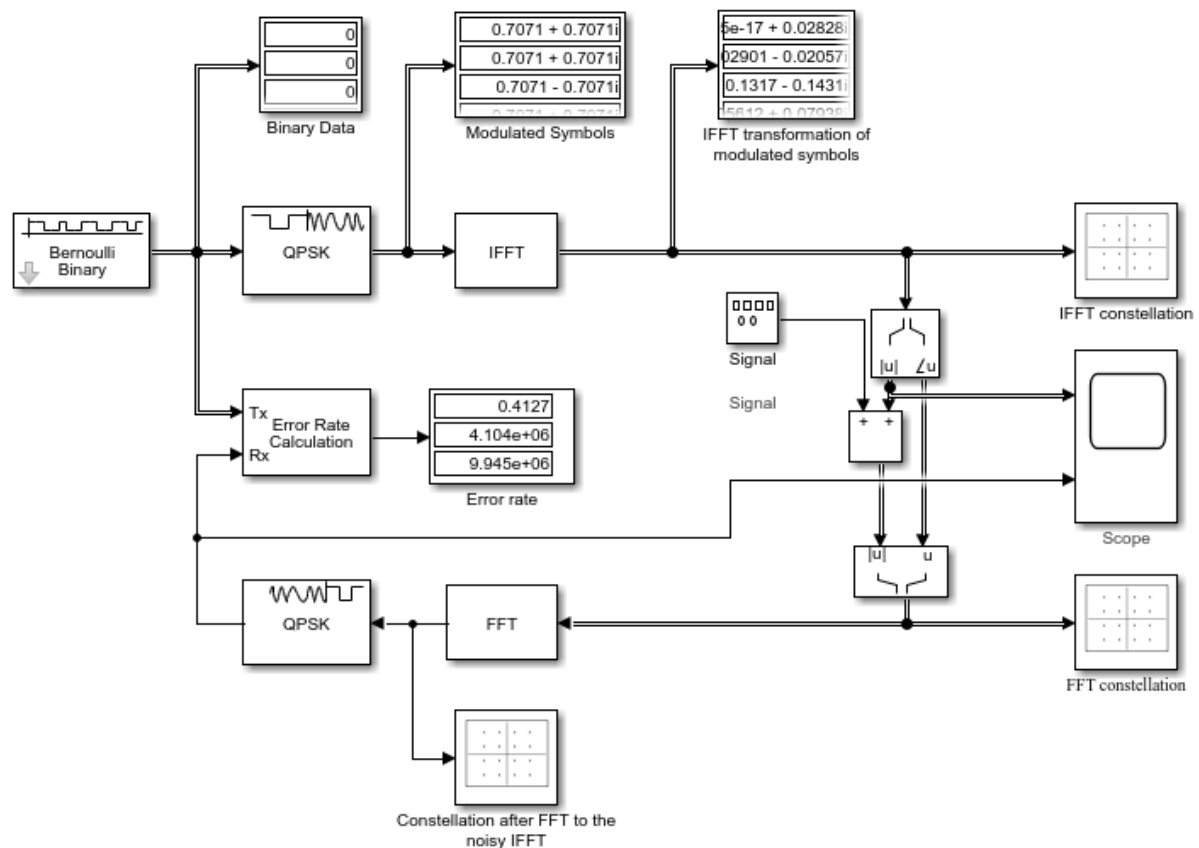
Aim: To perform OFDM Modulation and Demodulation.

Theory: Orthogonal Frequency Division Multiplexing (OFDM) is a type of multicarrier modulation technique in which higher data stream is divided into a number of lower data streams and transmitted over a large number of subcarriers. Various digital modulation techniques can be combined with OFDM for enhancing the performance of OFDM system. This paper focuses the MATLAB Simulink model of BPSK, QPSK and QAM based OFDM system which calculates the BER. A comparative study of bit error rate (BER) under normal AWGN channel has been done among various digital modulation techniques of OFDM using MATLAB Simulink model.



OFDM-modulation and demodulation

Fig. 10 a) OFDM Modulation and Demodulation

Simulink Model:**Fig. 10 b) Simulink Model of OFDM Modulation and Demodulation****Procedure:**

1. Connect the blocks as shown in the diagram.
1. For Bernoulli Binary block, choose source of initial seed as parameter, initial seed of 61, sample time as stime, samples per frame as 200 and output data type as double.
2. For QPSK, input type is bit, M array number is chosen as M and phase offset is chosen as π/M .
3. Chose random wave for signal.
4. For FFT and IFFT chose auto mode of implementation.

Output:

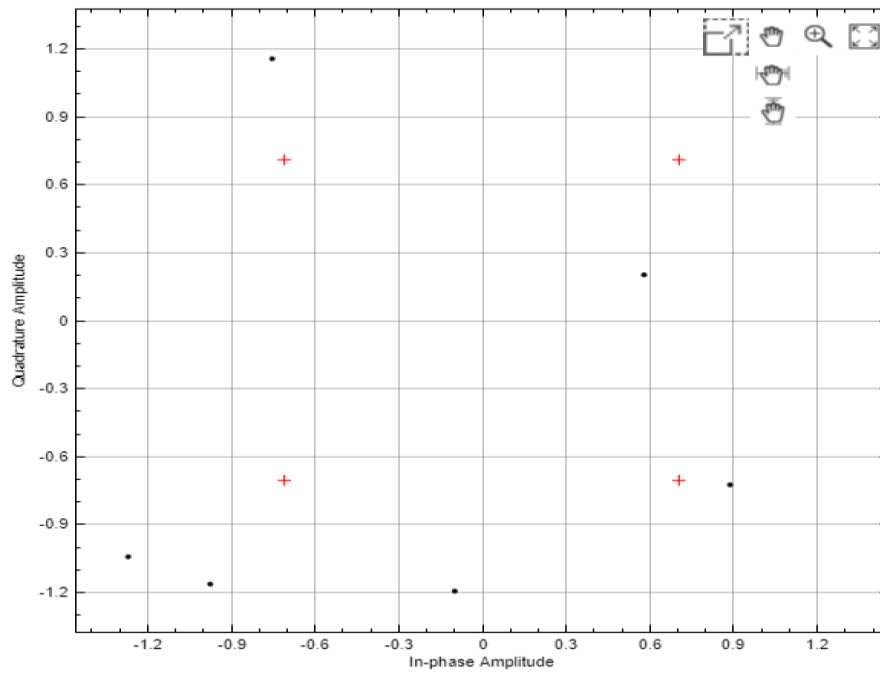


Fig. 10 c) Constellation after FFT to the noisy IFFT

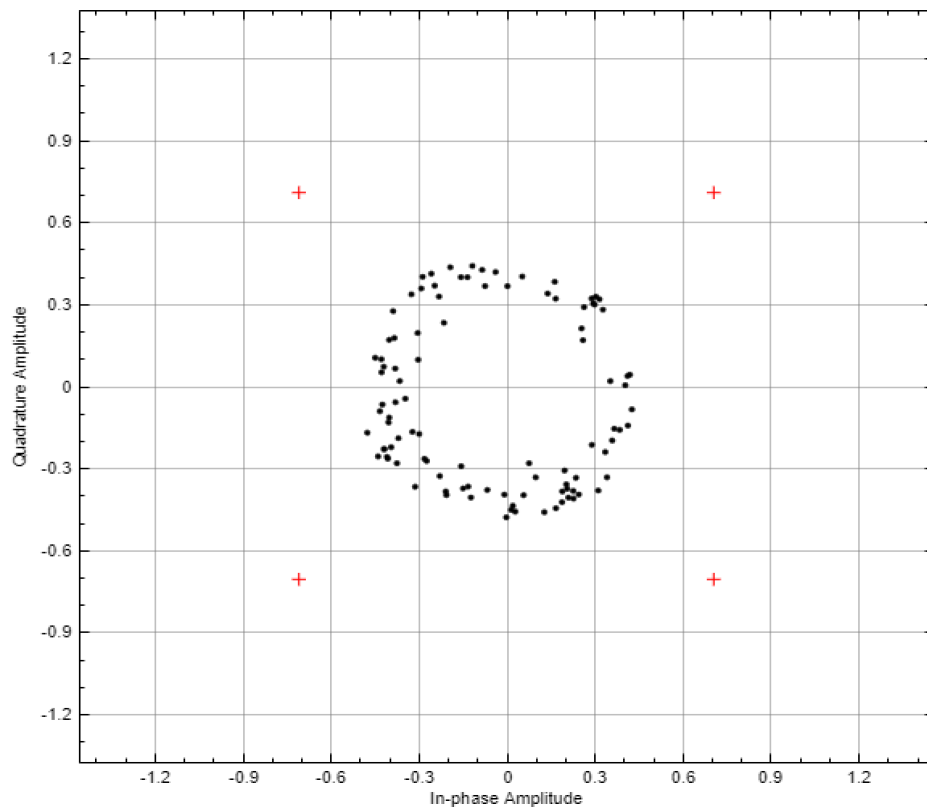


Fig. 10 d) FFT Constellation

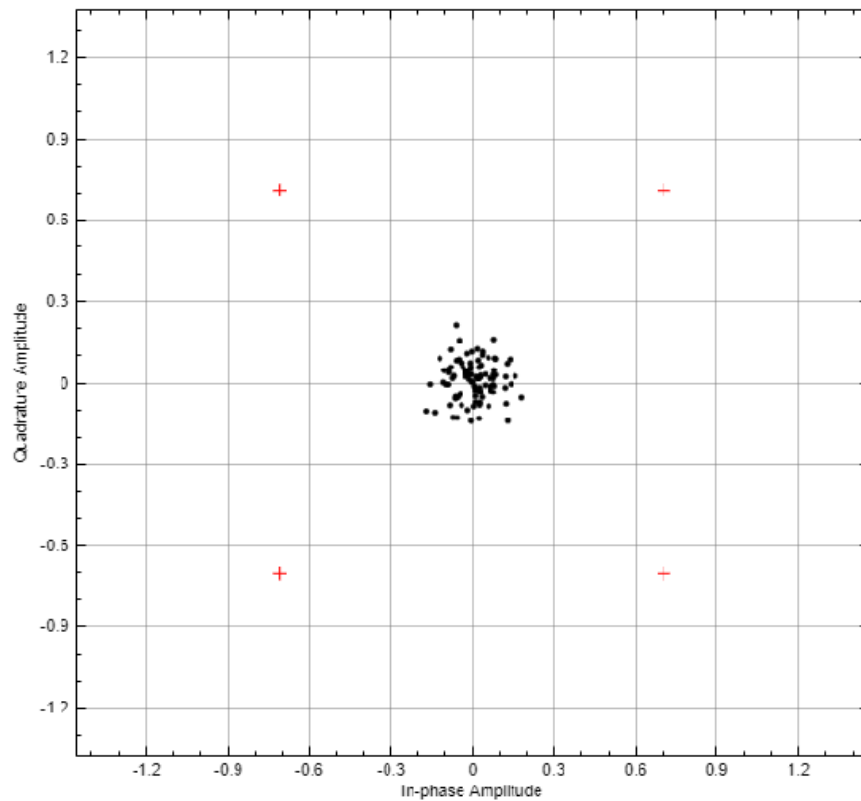


Fig. 10 e) IFFT Constellation

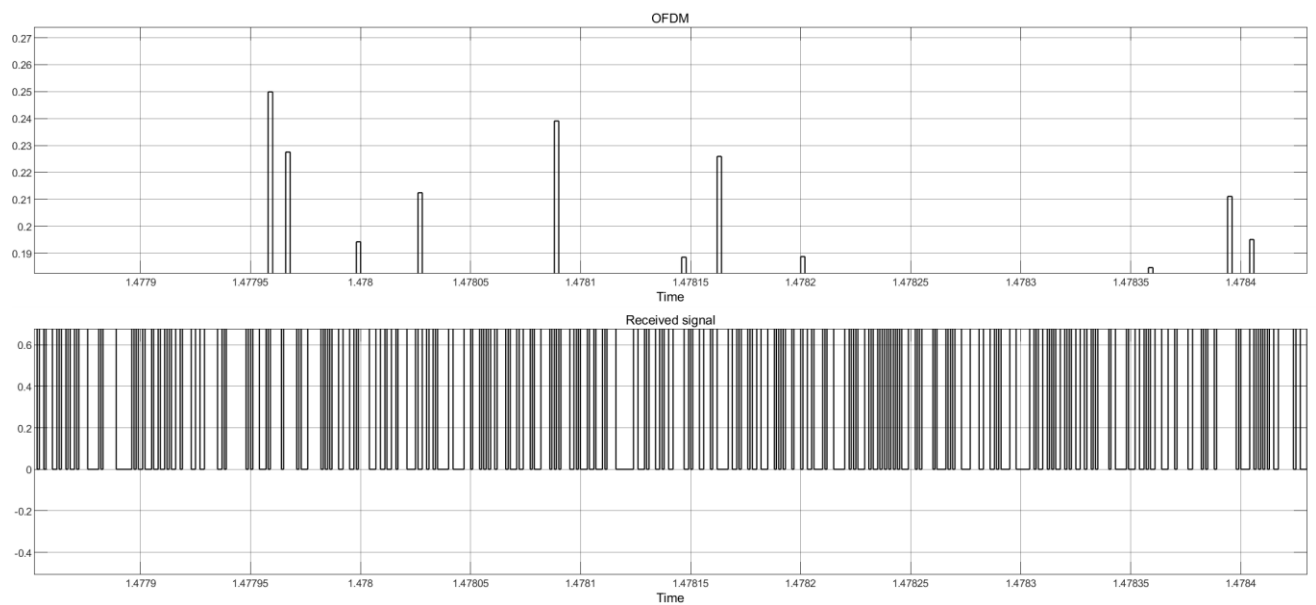


Fig. 10 f) OFDM and Received Signal

Result:

VIVA QUESTIONS

EXPERIMENT NO. 1 Generation of Basic Signals

1. What is Simulink, and how is it different from MATLAB?

Simulink is a graphical programming environment in MATLAB used for modeling, simulating, and analyzing dynamic systems. Unlike MATLAB, which is text-based, Simulink uses block diagrams for system representation.

2. What are the different types of basic signals that can be generated in Simulink?

The basic signals include:

- Sine Wave
- Square Wave
- Sawtooth Wave
- Step Signal
- Pulse Signal
- Ramp Signal

3. What is a sine wave?

Sine waves are continuous, periodic signals characterized by their smooth, oscillatory waveform. They are fundamental in analog communication systems, serving as carriers for transmitting information through modulation techniques such as Amplitude Modulation (AM), Frequency Modulation (FM), and Phase Modulation (PM).

4. How do you generate a sine wave in Simulink?

A sine wave can be generated using the Sine Wave block from the *Sources* library. The frequency, amplitude, phase, and sample time can be adjusted.

5. What parameters can be configured in the Sine Wave block?

The Sine Wave block allows configuring parameters like:

- Amplitude
- Frequency
- Phase
- Sample time

6. What is a square wave?

Square waves feature sharp changes in direction between the high and low discrete levels. They are frequently utilized in digital communication systems and are composed of odd harmonics. Square waves are employed as clock signals for synchronizing the transmission and reception of digital data. Additionally, they are employed in encoding processes using pulse width modulation (PWM) in digital-to-analog conversion.

7. What is a triangle wave?

Triangle waves are characterized by their linearly increasing and decreasing slopes, which combine to form a triangle. They are used in frequency modulation synthesis to create musical tones and in modulation/demodulation processes, particularly in phase-locked loop (PLL) circuits for frequency tracking and synchronization.

8. What is a ramp signal?

Ramp signals exhibit a linearly increasing or decreasing amplitude over time. They are used in communication systems for pulse shaping, where they shape transmitted pulses to meet specific bandwidth requirements and minimize distortion. Radar systems also use ramp signals to generate linear frequency modulation (LFM) waveforms for estimating target velocity and range.

9. What are the applications of sine wave and triangular wave?

Sine Wave

A sine wave is a fundamental waveform in signal processing and represents periodic oscillations.

Applications:

AC Power Supply: Alternating current (AC) follows a sine wave, making it essential in power distribution.

Communication Systems: Used in AM/FM radio, television signals, and audio processing.

Signal Processing: Acts as a basis for Fourier analysis in complex waveforms.

Mechanical Vibrations: Represents oscillatory motion in mechanical systems.

Medical Imaging: Used in ultrasound and MRI scanning.

Triangular Wave

A triangular wave is a non-sinusoidal waveform that increases and decreases *linearly*.

Applications:

Signal Modulation: Used in frequency modulation (FM) and function generators.

Analog Synthesizers: Produces sounds in electronic music.

Testing and Measurement: Employed in oscilloscopes and waveform analysis.

Pulse Width Modulation (PWM) Circuits: Controls switching in power electronics.

10. What are the applications of square wave and ramp signal?

Ramp Signal

A ramp signal is a continuous function that increases or decreases linearly over time.

Applications:

Control Systems: Used in system response analysis to test stability and behavior.

Oscilloscopes and Displays: Controls the horizontal sweep of cathode-ray oscilloscopes (CROs).

Motion Control: Represents uniform acceleration in robotics and mechanical systems.

Image Processing: Helps in intensity transformation and edge detection.

Square Wave

A square wave is a periodic waveform that alternates between two levels, typically high and low.

Applications:

Digital Electronics: Forms the basis of digital signals in computers and microprocessors.

Clock Signals: Provides timing signals for synchronous circuits.

Pulse Width Modulation (PWM): Used for motor speed control and dimming LED lights.

Communication Systems: Used in digital signal processing and clock synchronization.

Testing and Troubleshooting: Used in function generators for circuit testing.

EXPERIMENT NO. 2

Generation and Detection of DPSK Using Simulink

1. What is Differential Phase Shift Keying (DPSK)?

DPSK is a type of phase modulation where the information is encoded in the difference between consecutive phase changes rather than absolute phase values. This eliminates the need for a reference carrier at the receiver.

2. How does DPSK differ from PSK?

Unlike conventional PSK, where phase information is absolute, DPSK encodes data based on the phase difference between successive bits. This allows for non-coherent detection, making it simpler but slightly more error-prone than PSK.

3. Why is DPSK preferred over PSK in some applications?

- DPSK does not require a coherent carrier for demodulation, reducing system complexity.
- It is simpler to implement in practical communication systems.
- However, it has a slightly higher bit error rate (BER) compared to coherent PSK.

4. What is the role of a differential encoder in DPSK?

A differential encoder ensures that each bit is represented by the change in phase rather than an absolute phase value. This is crucial for DPSK modulation.

5. What is the main disadvantage of DPSK?

DPSK suffers from a higher BER compared to coherent PSK, especially in the presence of noise, since errors in one bit can propagate to the next bit.

6. How does a differential encoder work in DPSK modulation?

The differential encoder compares the current bit with the previous bit and modulates the phase accordingly:

- If the bit is 1, the phase remains the same.
- If the bit is 0, the phase is shifted by 180° .

7. What blocks are used in Simulink to generate DPSK?

The key blocks are:

- Bernoulli Binary Generator (for random bits)
- DPSK Modulator Baseband
- Scope Blocks (for signal visualization)

8. How does the carrier signal affect the DPSK modulation process?

DPSK does not use a separate carrier reference for detection. The phase change itself encodes the information, reducing the dependency on carrier synchronization.

9. What happens if the delay block is not used in DPSK demodulation?

The receiver will be unable to compare phase differences, leading to incorrect demodulation.

10. What are practical applications of DPSK?

- Wireless communication (Wi-Fi, Bluetooth)
- Satellite and optical communication
- RFID (Radio-Frequency Identification)
- Powerline communication systems

EXPERIMENT NO. 3

Generate a multiband signal using the Communications Toolbox

1. What is a multiband signal?

Multi-band signals are bandlimited signals that possess an additional structure in the frequency domain. The spectral support of a multi-band signal is restricted to several continuous intervals. Each of these intervals is called a band and it is assumed that no information resides outside the bands.

2. What is the significance of multiband signals?

- Efficient use of available spectrum.
- Used in multi-carrier modulation schemes such as OFDM.
- Allows parallel transmission of data in different frequency bands.

3. What are the applications of multiband signals?

- 5G and LTE communication.
- Wi-Fi and Bluetooth systems.
- Cognitive radio (dynamic spectrum allocation).
- Radar and satellite communication.

4. What happens if the frequency bands overlap?

Overlapping bands can cause inter-symbol interference (ISI) and distortion, degrading signal quality. Proper filtering and guard bands should be used to prevent this.

5. What is the role of filters in multiband signal generation?

Filters (such as bandpass filters) ensure that each frequency band remains within its designated range, preventing interference and distortion.

6. How does a multiband signal improve spectrum efficiency?

By using multiple frequency bands simultaneously, multiband signals increase data throughput and optimize spectral usage.

7. What are the challenges in generating and processing multiband signals?

- Synchronization issues between different bands.
- Inter-band interference if bands are too close.
- Complex receiver design for band separation.

8. How can you analyze the bandwidth of a multiband signal?

Using the Spectrum Analyzer block or the MATLAB function `spectrumAnalyzer`, we can examine the occupied frequency range.

9. What are the key components in the Communications Toolbox?

- Signal generation functions: Create individual baseband signals for each desired band using functions like `sin` or `sawtooth`.
- Frequency shifting: Use `comm.FrequencyShift` to shift each baseband signal to its desired carrier frequency within the designated band.
- `comm.MultibandCombiner`: This system object is crucial for combining the frequency-shifted signals into a single multiband signal by summing them together while ensuring proper sampling rates.

10. How do you combine multiple signals into a single multiband signal in Simulink?

The Sum Block is used to add different band-limited signals together, forming a multiband signal.

EXPERIMENT NO. 4
Random noise generation using Simulink & display
histogram plots of Gaussian and Uniform noise

1. What is random noise?
Random noise is an unpredictable signal that varies over time, commonly used in signal processing and communication systems to simulate real-world interference.
2. What are the types of random noise?
 - Gaussian noise (Normal noise) – Follows a normal distribution.
 - Uniform noise – Has a constant probability over a given range.
 - Impulse noise – Sudden, high-amplitude noise spikes.
 - White noise – Has equal power at all frequencies.
3. What is the significance of random noise in communication systems?
 - Simulates real-world interference (e.g., thermal noise, atmospheric noise).
 - Helps test signal robustness in noisy environments.
 - Used in filter design and performance evaluation.
4. How do you display the histogram plot of Gaussian and Uniform noise in Simulink?
 - Generate noise using the Random Number or Uniform Random Number block.
 - Connect the output to a Histogram Block.
 - Run the simulation and observe the histogram distribution.
5. What is the purpose of the histogram in noise analysis?
A histogram shows the probability distribution of noise values, helping to verify whether the generated noise follows the expected distribution (Gaussian or Uniform).
6. What is Gaussian noise?
Gaussian noise (or normal noise) has a bell-shaped probability distribution, where most values are near the mean, and extreme values are less frequent.
7. Where is Gaussian noise commonly found?
 - Thermal noise in electrical circuits.
 - Channel noise in wireless communication.
 - Image processing (e.g., camera sensor noise).
8. What is uniform noise?
Uniform noise has a constant probability for all values within a specific range $[a,b]$, meaning every noise value is equally likely.
9. Where is uniform noise used?
 - Testing systems with random input signals.
 - Cryptography (random number generation).
 - Monte Carlo simulations.
10. What is the difference between white noise and Gaussian noise?
 - White noise has equal power across all frequencies.
 - Gaussian noise refers to noise with a Gaussian amplitude distribution, which may or may not be white.

EXPERIMENT NO. 5

Perform Equalization, Convolution, and Cyclic Prefix Addition on basic OFDM

1. Define Orthogonal Frequency Division Multiplexing (OFDM).
Orthogonal Frequency Division Multiplexing (OFDM) is a high-efficiency modulation technique used in modern communication systems, such as Wi-Fi, LTE, and 5G, to transmit large amounts of data over radio waves.
2. How OFDM is used to combat inter-symbol interference (ISI)?
OFDM helps to combat ISI through the following mechanisms:
 - Cyclic Prefix (CP)
 - Symbol Duration and Guard Interval
 - Parallel Transmission and Low Data Rate on Each Subcarrier
 - Robustness to Frequency-Selective Fading
3. What is the role of the inverse Fast Fourier Transform (IFFT) in OFDM systems?
 - Transformation from Frequency Domain to Time Domain
 - Simultaneous Transmission on Multiple Subcarriers
 - Orthogonality of Subcarriers
 - Creation of the OFDM Signal
4. What is the purpose of equalization in an OFDM system, and why is it necessary?
 - Multipath Propagation and Frequency-Selective Fading
 - Inter-Symbol Interference
 - Channel Distortion
5. What are the different types of equalizers used in OFDM.
 - Zero-Forcing (ZF) Equalizer
 - Minimum Mean Square Error (MMSE) Equalizer
 - Decision Feedback Equalizer (DFE)
 - Adaptive Equalizer
 - Frequency Domain Equalizer (FDE)
6. What is the difference between a frequency-domain equalizer and a time-domain equalizer?
 - Frequency-domain equalizers are more commonly used in OFDM systems where the received signal is transformed into the frequency domain using FFT. They efficiently handle frequency-selective fading and are generally simpler to implement.
 - Time-domain equalizers process the signal in the time domain and are particularly effective in mitigating Inter-Symbol Interference (ISI) and compensating for multipath propagation. They are more complex but are necessary in scenarios with time-varying distortions (e.g., Doppler effects or rapid fading).
7. Define Minimum Mean Square Error (MMSE) equalizer.
The Minimum Mean Square Error (MMSE) equalizer is a type of linear equalizer commonly used in communication systems to minimize the error between the transmitted signal and the received signal, considering both channel distortion and noise.
8. What role does convolution play in the context of OFDM systems?
 - Multipath propagation causes convolution of the transmitted signal with the channel's impulse response, resulting in distortions like ISI and ICI.

- Cyclic Prefix (CP) in OFDM transforms this linear convolution into a circular convolution, making it easier to manage and mitigate ISI.
- Channel estimation and equalization techniques are used to reverse the effects of the convolution, improving the signal recovery process.

9. What is the difference between convolutional coding and block coding?

- Convolutional coding is generally preferred in systems where real-time transmission is critical, and error correction needs to be more robust against burst errors and noisy channels.
- Block coding is simpler to implement and is useful in systems where data is transmitted in fixed-size blocks, such as data storage or error detection applications.

10. What is a cyclic prefix (CP), and why is it added to each OFDM symbol?

- Cyclic Prefix (CP) is a copy of the last portion of the OFDM symbol that is appended to the front of the symbol before transmission.
- The cyclic prefix provides a guard interval between symbols, preventing them from interfering with each other, and enables simpler signal processing techniques like the FFT at the receiver.

EXPERIMENT NO. 6

QPSK Transmitter and Receiver in Simulink

1. What is QPSK (Quadrature Phase Shift Keying)?
QPSK is a type of phase modulation where two bits are encoded into each symbol. In QPSK, there are four possible phase states (0° , 90° , 180° , and 270°) corresponding to two bits per symbol. This allows QPSK to transmit twice the data rate of BPSK (Binary Phase Shift Keying) with the same bandwidth.
2. What is the basic principle of operation of a QPSK transmitter in Simulink?
A QPSK transmitter in Simulink generally consists of the following stages:
 - Bit Source
 - Bit-to-Symbol Mapping
 - In-phase (I) and Quadrature (Q) Components
 - Carrier Signals
 - Summing the Signals
 - Transmission
3. What is the role of the I (In-phase) and Q (Quadrature) components in QPSK?
In QPSK, the I (In-phase) and Q (Quadrature) components represent the two components of the modulated signal that carry information. The I and Q components are orthogonal to each other (90° phase shift). Together, they allow for the encoding of two bits of data per symbol.
4. How can you implement QPSK modulation in Simulink?
 - Bit Generator
 - Reshape the Bit Stream
 - Symbol Mapping
 - I and Q Channels
 - Carrier Modulation
 - Summing
5. What is the QPSK receiver structure in Simulink?
A typical QPSK receiver in Simulink consists of the following steps:
 - Received Signal
 - Matched Filtering
 - Demodulation
 - Symbol Decision
 - Bit Mapping
 - Error Detection
6. How do you handle noise in a QPSK communication system in Simulink?
 - AWGN Channel
 - Matched Filtering
 - Noise Impact on Bit Error Rate (BER)
7. What are some common errors encountered in QPSK communication?
 - Inter-Symbol Interference (ISI)

- Phase Noise
 - Bit Errors
8. How can you calculate the Bit Error Rate (BER) in a QPSK system in Simulink?
BER= Number of Errors/Total Number of Bits Transmitted
9. What are the advantages of QPSK over BPSK?
- Higher Data Rate
 - Spectral Efficiency
 - Bandwidth Utilization
10. How can you visualize the QPSK signal and constellation in Simulink?
- Scope
 - Constellation Diagram
 - Scatter Plot

EXPERIMENT NO. 7

Design and Implementation of Low Pass and High Pass filter

1. What is the difference between a low-pass and high-pass filter?
 - Low-pass filter (LPF): A low-pass filter allows signals with frequencies below a certain cutoff frequency to pass through and attenuates frequencies above the cutoff. It is typically used to filter out high-frequency noise or unwanted signals.
 - High-pass filter (HPF): A high-pass filter allows signals with frequencies above a certain cutoff frequency to pass through and attenuates frequencies below the cutoff. It is used to filter out low-frequency noise or unwanted components, such as DC signals.
2. How is a low-pass filter designed?

LPF Can be designed using algorithms such as the Butterworth, Chebyshev, or Elliptic filters, depending on the desired characteristics like passband ripple or stopband attenuation.
3. How is a high-pass filter designed?

High-pass filter can be designed in the digital domain using algorithms such as FIR (Finite Impulse Response) or IIR (Infinite Impulse Response) filters with the desired cutoff frequency and filter characteristics.
4. What is the significance of the cutoff frequency in both low-pass and high-pass filters?
 - The cutoff frequency is the frequency at which the filter begins to attenuate the signal. It defines the boundary between the passband and the stopband of the filter.
 - At the cutoff frequency, the filter's output signal power is reduced by half (a -3 dB drop).
5. Explain the frequency response of a low-pass filter?

For a low-pass filter, frequencies below the cutoff pass through, while frequencies above the cutoff are attenuated.
6. Explain the frequency response of a high-pass filter?

For a high-pass filter, frequencies above the cutoff pass through, while frequencies below the cutoff are attenuated.
7. What are the key differences between analog and digital filters?

Analog Filters:

 - Process continuous-time signals.
 - Implemented using components like resistors, capacitors, inductors, and operational amplifiers.
 - Filters like RC, RLC, and active filters are common examples.
 - Characteristics such as cutoff frequency, bandwidth, and filter type (low-pass, high-pass, etc.) are determined by the component values.

Digital Filters:

 - Process discrete-time signals.
 - Implemented through algorithms using discrete-time systems and can be simulated or implemented on microcontrollers, DSP chips, or software.
 - Commonly used for precise control over filtering characteristics and can be implemented using FIR or IIR filters.

- Digital filters allow for more flexibility, like adjustable parameters and better accuracy in performance.
8. What are the advantages of using a Butterworth filter?
 - Maximally Flat Response
 - Smooth Roll-off
 - Simplicity
 9. How does the filter order affect the filter performance?
 - Higher Filter Order: A higher filter order results in a steeper roll-off at the cutoff frequency, meaning the filter will more sharply separate the passband from the stopband. However, a higher order also means more components or more complexity in a digital filter, leading to higher computational requirements and possibly more delay.
 - Lower Filter Order: A lower order results in a more gradual roll-off, but the filter may not effectively attenuate unwanted frequencies that are far from the cutoff. The filter will be simpler and computationally cheaper.
 10. How can you calculate the filter order for a given specification?

The order of a filter determines the complexity and the steepness of the filter's frequency response. For a low-pass or high-pass filter, the order can be calculated using formulas based on the desired cutoff frequency, passband ripple, and stopband attenuation.

EXPERIMENT NO. 8

Design and Implementation of Band Pass and Band Reject filter

1. What is a band-pass filter?
A band-pass filter allows frequencies within a specific range (bandwidth) to pass while attenuating frequencies outside this range.
2. What is a band-reject (notch) filter?
A band-reject filter attenuates a specific range of frequencies while allowing frequencies outside this range to pass through.
3. What are the applications of band-pass and band-reject filters?
 - Band-pass filters: Audio processing, wireless communication, biomedical applications (EEG, ECG).
 - Band-reject filters: Noise cancellation, power line interference suppression (e.g., removing 50Hz hum in audio circuits).
4. What is the difference between active and passive filters?
 - Active filters use active components (op-amps, transistors) and do not require inductors. They can amplify signals.
 - Passive filters use only resistors, capacitors, and inductors and do not amplify signals.
5. What are the advantages of active filters over passive filters?
 - No need for bulky inductors.
 - Can provide gain and better impedance matching.
 - More stable and easier to design.
6. How do you design a band-pass filter?
A band-pass filter can be designed using a combination of a low-pass filter and a high-pass filter, ensuring their cutoff frequencies define the passband.
7. How do you design a band-reject filter?
A band-reject filter is designed by combining a low-pass and high-pass filter in parallel, where their cutoff frequencies define the rejected band.
8. What is the significance of cutoff frequencies in these filters?
The cutoff frequencies define the range where attenuation begins. In band-pass filters, they define the bandwidth. In band-reject filters, they define the rejected frequency range.
9. What are the key components used in active band-pass and band-reject filters?
Resistors (R), capacitors (C), and operational amplifiers (op-amps).
10. What is the significance of gain in a band-pass filter?
The gain determines the amplification of signals within the passband. Higher gain results in stronger output signals.

EXPERIMENT NO. 9

Obtain the scatter plots & Eye diagrams of a QPSK signal to visualize the signal behavior in presence of AWGN

1. What is QPSK?
Quadrature Phase Shift Keying (QPSK) is a digital modulation scheme where data is transmitted by shifting the phase of a carrier signal among four possible values (0° , 90° , 180° , and 270°).
2. What is a scatter plot in the context of QPSK?
A scatter plot is a graphical representation of received signal points (constellation points) in the in-phase (I) and quadrature (Q) plane, helping to visualize the modulation and noise effects.
3. What is an eye diagram?
An eye diagram is a tool used to evaluate the quality of a digital signal. It overlays multiple segments of a signal waveform, showing signal integrity, noise effects, and timing errors.
4. What is AWGN?
Additive White Gaussian Noise (AWGN) is a type of noise with a constant power spectral density across all frequencies and follows a Gaussian distribution in amplitude.
5. How does AWGN affect a QPSK signal?
AWGN introduces random variations in phase and amplitude, causing distortion in the constellation diagram and signal degradation in the eye diagram.
6. How does noise affect the scatter plot of a QPSK signal?
In the presence of noise, the constellation points spread around their ideal locations, forming a cloud-like shape instead of precise points.
7. What happens to the eye diagram when noise is introduced?
The eye opening starts to close, indicating signal distortion and reduced signal integrity. A completely closed eye means severe degradation.
8. How does signal-to-noise ratio (SNR) impact the scatter plot and eye diagram?
Higher SNR results in well-defined constellation points and a wider eye opening. Lower SNR causes more spread in the scatter plot and a closing eye diagram.
9. Why is QPSK preferred over BPSK in many applications?
QPSK transmits two bits per symbol, effectively doubling the data rate compared to BPSK without increasing bandwidth requirements.
10. What is the significance of symbol error rate (SER) in QPSK?
SER measures the probability of incorrect symbol detection. It increases as noise level increases.

EXPERIMENT 10

Perform OFDM Modulation and Demodulation

1. What is OFDM?
Orthogonal Frequency Division Multiplexing (OFDM) is a digital modulation technique that divides a data stream into multiple parallel subcarriers, each modulated at a lower data rate.
2. Why is OFDM used in communication systems?
OFDM improves spectral efficiency, reduces inter-symbol interference (ISI), and provides robustness against multipath fading.
3. What are the key applications of OFDM?
OFDM is widely used in Wi-Fi (802.11a/g/n/ac/ax), LTE, 5G, DVB-T, ADSL, and underwater communications.
4. What is the main advantage of OFDM over traditional single-carrier modulation?
OFDM effectively combats multipath fading and ISI by dividing data across multiple orthogonal subcarriers.
5. What is orthogonality in OFDM?
Orthogonality ensures that subcarriers do not interfere with each other, allowing efficient bandwidth utilization without inter-carrier interference (ICI).
6. How is OFDM modulation performed?
OFDM modulation involves:
 - Serial-to-parallel conversion
 - Mapping bits to symbols (QPSK, QAM)
 - Applying Inverse Fast Fourier Transform (IFFT)
 - Adding a cyclic prefix (CP)
 - Parallel-to-serial conversion for transmission
7. What is inter-symbol interference (ISI) and how does OFDM handle it?
ISI occurs when one symbol overlaps with another due to multipath. OFDM minimizes ISI using a cyclic prefix and maintaining orthogonality.
8. What is the role of IFFT and FFT in OFDM?
IFFT converts frequency-domain subcarriers into a time-domain signal for transmission. FFT at the receiver transforms the received signal back into the frequency domain for decoding.
9. What is the significance of a cyclic prefix (CP) in OFDM?
The cyclic prefix acts as a guard interval to combat ISI by ensuring that delayed versions of a signal do not interfere with the next symbol.
10. What happens if the cyclic prefix is too short?
A short cyclic prefix may not fully eliminate ISI, causing signal distortion and degraded system performance.

Institute Vision

Become a premier institution imparting quality education in engineering and management to meet the changing needs of society.

Institute Mission

- M1.** Create environment conducive for continuous learning through quality teaching and learning processes supported by modern infrastructure.
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- M3.** Inculcate ethical values and environmental consciousness through holistic education programs.

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To provide Quality technical education in Electronics and Communication Engineering focus on global challenges and societal needs.

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